

TopK Observables as Projection-Selection Readbacks of Hidden-State Geometry

A Conditional-Mean Analysis of Vocabulary-Side Token Sets

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Abstract

Identity-only TopK sets provide a natural vocabulary-side view of language model hidden states. When logits, probabilities, rank scores, and within-set weights are removed, the remaining token identities still preserve substantial hidden-state neighborhood geometry. This effect initially appears to suggest that learned vocabulary geometry preserves semantic row-identity structure. We show that the stronger semantic attribution is not supported.

We study identity-only TopK sets as geometry-preserving observables and audit the source of their preservation. Corrected W -disruption experiments show that randomized, permuted, and matched Gaussian readout matrices can match or exceed the real vocabulary readout matrix. Row permutation does not materially reduce preservation, and matched Gaussian nulls reach near-ceiling performance. Thus, topology preservation cannot be attributed directly to semantic row-identity structure in W .

We identify the underlying mechanism as **projection-selection conditional-mean readback**. For query vectors H_i and candidate readout vectors R_j , TopK selection by $H_i R_j^\top$ conditions the selected vectors on extreme projection along H_i . Their centroid is therefore shifted toward the query direction:

$$Z_i = \frac{1}{k} \sum_{j \in \text{TopK}(H_i R^\top)} R_j = \alpha(d, V, k) H_i + \epsilon_i.$$

For Gaussian candidates, the empirical α matches the conditional-mean prediction with very small error. BottomK selection recovers the opposite query direction, while random-K selection carries little structure. The preservation of pairwise geometry follows from this noisy query-direction estimator.

These results revise the interpretation of TopK observables. Identity-only TopK sets are valid readout-side observables of hidden-state geometry, but their topology-preservation effect is largely explained by a general high-dimensional projection-selection mechanism. Learned W may still provide semantic content and distributional modulation, but topology preservation alone is not evidence for learned semantic row-identity causality.

1. Introduction

The starting point of this work is a modest observation. Large language models are difficult to interpret, but difficulty does not imply that their internal computation is intrinsically inaccessible. A more useful suspicion is that some commonly measured objects are too close to the implementation layer. Neurons, activation vectors, and parameters are measurable, yet they are not automatically the right objects for describing the

mathematical structure of inference.

Recent topological studies of hidden representations suggest a different coordinate system. Instead of treating a hidden state only as a high-dimensional vector, they ask whether collections of hidden states carry recoverable topology. This shift is useful because topology is less tied to individual neuron axes. It asks whether model computation has a structural shape that survives changes in representation.

A remaining gap then appears. Hidden-state topology gives a mathematical object, but it does not by itself explain how that object reaches the model's vocabulary-level generative interface. The question is not only whether hidden states have structure. The more specific question is whether some vocabulary-side observable preserves enough of that structure to be useful.

The vocabulary readout matrix W offers a natural place to look. During generation, a hidden state is read through W into vocabulary space. The same readout can also be applied at intermediate layers. At each layer, $H_l W^\top$ induces a ranked set of vocabulary tokens. The first empirical question is whether the identity-only TopK set induced by this readout preserves hidden-state neighborhood geometry after scores are discarded.

The answer is yes. Broad identity-only TopK sets preserve hidden-state geometry well above random token sets and several token-level controls. But this positive result creates a second, harder question: why does it work? The initial semantic reading would be that learned vocabulary geometry in W exposes a readout-side TopK observable of the hidden state. That reading is too strong.

A stronger geometric mechanism is present. If a query vector H_i is projected onto many high-dimensional candidate readout vectors R_j , and the largest entries of $H_i R_j^\top$ are selected, the centroid of the selected vectors estimates the query direction. In simplified form,

$$Z_i = \frac{1}{k} \sum_{j \in \text{TopK}(H_i R^\top)} R_j = \alpha(d, V, k) H_i + \epsilon_i.$$

The opposite extreme has the corresponding sign reversal:

$$Z_i^{\text{bottom}} \approx -\alpha(d, V, k) H_i + \epsilon_i.$$

Since global sign reversal preserves cosine neighborhood geometry, both TopK and BottomK can preserve pairwise query structure. Random-K selection cannot. This is the projection-selection conditional-mean readback mechanism studied in this paper.

The object studied here is deliberately narrow. We define the identity-only TopK set as

$$S_k(H_l) = \text{TopK}_k(H_l W^\top),$$

and the selected embedding set as

$$\mathcal{E}_k(H_l) = \{w_i : i \in S_k(H_l)\}.$$

The important point is that this is not hidden-space kNN. Hidden-kNN constructs neighborhoods among hidden vectors and is used to reconstruct topology directly in hidden space. The object above is different: it is a vocabulary-induced readout set selected by a hidden state through the model head.

This paper makes three contributions. First, we show that identity-only TopK sets preserve hidden-state geometry above random and shuffled token-set baselines. Second, we show that this preservation is not specific to learned vocabulary geometry: W -disruption, row-permutation, and matched Gaussian controls demonstrate that the effect survives without semantic row identity and can be matched or exceeded by high-

dimensional random readout matrices. Third, we identify the mechanism behind the effect: TopK selection over high-dimensional projections induces a conditional-mean shift of selected readout vectors along the query direction. The centroid of the selected set therefore acts as a noisy estimator of the hidden query direction.

The claim is intentionally bounded. This paper does not claim that TopK causes inference, that W is a clean semantic manifold, or that topology preservation proves semantic row-identity structure in W . It establishes and explains a readout phenomenon: identity-only TopK sets preserve hidden-state geometry, and this preservation is primarily explained by projection-selection conditional-mean readback. Any stronger semantic claim must be tested as a residual beyond a matched projection-selection null.

2. Related Work

2.1 Hidden-State Topology and kNN Reconstruction

Recent work in topological data analysis has begun to examine the topology of neural and language-model hidden states. A common pipeline constructs neighborhood graphs or filtrations over hidden-state point clouds and then uses persistent homology or zigzag persistence to track topological structure across layers. This line of work motivates the use of local-neighborhood constructions as tools for recovering hidden-state topology.

This work does not replicate hidden-space kNN. Instead, it asks whether a different object, the identity-only TopK set, inherits enough hidden-state neighborhood geometry to serve as a vocabulary-side geometric observable.

This external topological literature is best understood here as a hidden-space coordinate system. It shows that hidden states can have measurable topology, but it does not by itself explain how that topology is exposed at the model's vocabulary interface. The paper tests whether identity-only TopK set supplies such an interface. In this sense, hidden-space TDA provides the topology-reconstruction coordinate system, while identity-only TopK set provides a candidate vocabulary-side observation surface.

2.2 Vocabulary Embeddings, Output Geometry, and Readout Bias

This paper initially treated the vocabulary readout matrix W as a learned geometric substrate on which hidden states induce local neighborhoods. That motivation remains useful, but the projection-selection geometry audits require a sharper distinction. Learned vocabulary embeddings can contain semantic organization, frequency structure, anisotropy, hubs, row-norm variation, tokenization artifacts, and corpus-induced covariance. A positive TopK preservation result cannot by itself identify which of these components is responsible.

The key correction is that readout geometry alone can preserve hidden-state geometry. Even random Gaussian readout matrices can produce strong TopK conditional-mean readback when selection is performed by $H_i R^\top$. Therefore, the empirical question is no longer whether TopK subsets preserve topology. They do. The harder question is whether real learned W contributes any residual beyond matched projection-selection conditional-mean readback nulls.

This revised position is more conservative than a semantic-row-identity claim. The vocabulary matrix supplies the interface through which identity-only TopK sets are observed, but topology preservation must be decomposed into a general projection-selection readback component and any remaining learned- W residual.

2.3 Representation Similarity and Proxy-Topological Diagnostics

This work uses representation-similarity tools to compare the geometry of hidden states and TopK-derived signatures. Distance Spearman correlation measures pairwise distance-order preservation. KNN overlap measures local neighborhood preservation. CKA measures global representational alignment. These metrics do not replace persistent homology, but they provide a practical first audit of whether the TopK-derived representation preserves hidden-state geometry.

2.4 Logit Lens, Tuned Lens, and Vocabulary-Space Probes

This paper is also related to vocabulary-space probing methods such as the logit lens and tuned lens. The logit lens applies the model's vocabulary readout to intermediate hidden states, while the tuned lens trains layer-specific affine probes to decode intermediate representations into vocabulary distributions. These methods ask what prediction-like information is visible at each layer.

The present object is different. We do not analyze the top predicted token, the probability distribution, logit values, rank gaps, or layerwise prediction trajectories. Instead, after projecting an intermediate hidden state through the vocabulary matrix, we discard logit magnitudes, probabilities, and rank order, keep only the identity of the selected TopK token set, and ask whether this identity-only set preserves hidden-state neighborhood geometry. So, logit-lens-style work motivates vocabulary-space observation, while this paper tests whether a score-free token set can serve as a vocabulary-side geometric observable.

3. Object: Identity-Only TopK Sets as Projection-Selection-Biased Readout Observables

The object studied in this paper should not be understood as a static Prompt-conditioned readout-side TopK observable of hidden space. That interpretation would be too strong, and it would also miss the role of the input prefix. The TopK set is not simply a coordinate patch around a hidden vector. It is a vocabulary-level selected set induced by a hidden state after that hidden state has already been contextualized by the prompt.

3.1 Hidden-kNN and Vocabulary-Induced TopK Are Different Objects

The key object in this paper is easy to confuse with hidden-space kNN, so the distinction must be made explicit.

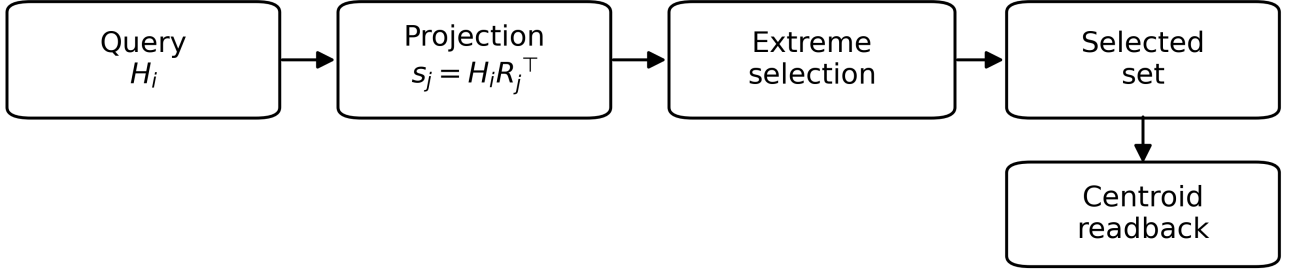
A hidden-space kNN construction builds a selected set graph among hidden vectors. Given a hidden point cloud $\{H_l(x_i)\}_{i=1}^n$, it asks which hidden states are near each other in representation space. This object is useful for topology reconstruction.

identity-only TopK set begins with a single contextualized hidden state. It projects that state through the vocabulary matrix W and selects the tokens with the highest logits. It therefore asks a different question: which vocabulary tokens does this hidden state most strongly activate through the model head?

The two objects share a locality principle, but they live in different spaces.

Object	Construction	Space	Main use
hidden-kNN	$kNN_k(h)$ over hidden vectors	hidden representation space	topology reconstruction

Object	Construction	Space	Main use
identity-only TopK set	$\text{TopK}_k(H_l W^\top)$	readout-indexed token set	score-free readout observation



$$\text{TopK}: \mathbb{E}[R \mid H_i R^\top \geq t] = +\alpha H_i$$

$$\text{BottomK}: \mathbb{E}[R \mid H_i R^\top \leq -t] = -\alpha H_i$$

Selected-set centroids are noisy conditional-mean estimators of the query direction.

Figure 1. Projection-selection conditional-mean readback. A query vector induces projection scores over candidate readout vectors. Extreme selection shifts the selected-vector centroid toward the query direction for TopK and toward the opposite direction for BottomK. This mechanism is the central explanation tested in the paper.

The bridge tested in this paper is not

$$\text{TopK}_{\text{hidden}} = \text{TopK}_{\text{vocab}}.$$

The bridge is weaker and more useful:

$$\text{Topology}(H_l) \rightarrow \text{Topology}(\mathcal{E}_k(H_l)).$$

3.2 Definition

Given an intermediate hidden state $H_{l,t}$ and the vocabulary readout matrix W , we define the intermediate vocabulary readout as

$$P_{l,t} = H_{l,t} W^\top.$$

The identity-only TopK set is

$$S_k(H_{l,t}) = \{i \in V : P_{l,t,i} \text{ is among the top } k \text{ entries of } P_{l,t}\}.$$

The corresponding selected embedding set is

$$\mathcal{E}_k(H_{l,t}) = \{w_i \in W : i \in S_k(H_{l,t})\}.$$

Here $S_k(H_{l,t})$ is the identity-level representation of the selected set: it contains only token indices. The selected embedding set $\mathcal{E}_k(H_{l,t})$ maps those indices back into the vocabulary geometry through their associated vectors w_i .

When no ambiguity arises, we use $S_k(H_l)$ for the identity-only TopK set and $\mathcal{E}_k(H_l)$ for its selected embedding set. We avoid calling either object a semantic row-identity structure. All geometric signatures are computed from $\mathcal{E}_k(H_l)$.

3.3 Readout-Side Geometric Observable with projection-selection geometry Bias

The object should not be described as direct evidence for a learned semantic mechanism in W . Corrected W -disruption and projection-selection geometry audits make that reading too broad. Row-permuted, random-Gaussian, and orthogonal-rotated readout constructions can preserve substantial hidden-state neighborhood geometry. Pure Gaussian readouts can do the same without any model, tokenizer, or learned vocabulary matrix.

The narrower object still matters. A TopK token set is a discrete vocabulary-side trace of a hidden state. It is not a random token bag, and it is not the same as the input tokens. It carries enough structure to reconstruct hidden-state neighborhood relations under proxy-topological diagnostics. What the present paper establishes is this observable status and the projection-selection mechanism behind it.

The semantic contribution of W is not supported by topology preservation alone. the real-readout residual audit finds no positive topology-preservation residual of real W over matched projection-selection nulls, and row permutation does not materially reduce preservation. The safer term is therefore projection-selection-biased readout-side geometric observable: the set is induced through the vocabulary interface, measured in vocabulary geometry, and informative about hidden-state geometry, but its preservation is primarily explained by high-dimensional projection-selection conditional-mean readback.

3.4 Why Identity-Only TopK?

There is an obvious weak version of the hypothesis that should not be overvalued. The full logit vector $H_l W^\top$ is a linear readout of the hidden state, so it is expected to retain information about the hidden representation. A positive result at the full-logit level would therefore be difficult to interpret.

The harder question is whether the identity set induced by the ordering of these logits also preserves structure. The answer is yes, but projection-selection geometry changes the interpretation of that answer. The TopK set is not only a score-free token object; it is also the output of an extreme-selection operation over high-dimensional projections. Once the highest-scoring vectors are selected, their centroid can estimate the query direction. Thus, identity-only TopK removes logit magnitudes and probabilities, but it does not remove the geometric bias introduced by selection.

This is exactly why the identity-only test is useful. It exposes a nontrivial observable while also revealing a strong confound. We remove logit magnitudes, probabilities, and rank scores, and keep only the identities of the selected tokens:

$$S_k(H_l) = \text{TopK}_k(H_l W^\top).$$

The question is then: how much hidden geometry is carried by the selected set, and how much of that preservation is explained by projection-selection readback rather than learned vocabulary semantics?

4. Signatures, Metrics, and Experimental Setup

4.1 Identity-Only Signatures

We construct three identity-only signatures of the identity-only TopK set.

Center

The first signature is the selected-set centroid:

$$C_k(H_{l,t}) = \frac{1}{k} \sum_{i \in S_k(H_{l,t})} w_i.$$

This center asks a minimal question: if only the selected token identities are retained, does the centroid of their vocabulary vectors preserve information about the hidden state that selected them?

Center + Spread

The second signature augments the center with a dispersion term. Define

$$r_k(H_{l,t}) = \frac{1}{k} \sum_{i \in S_k(H_{l,t})} (1 - \cos(w_i, C_k(H_{l,t}))).$$

The center+spread signature is

$$Z_k^{cs}(H_{l,t}) = (C_k(H_{l,t}), r_k(H_{l,t})).$$

The spread term is included because two identity-only TopK sets may have similar centers but different selected-set dispersion. A compact selected set and a diffuse selected set can point in roughly the same direction while encoding different degrees of local uncertainty or semantic breadth.

Center + Spectrum

The third signature adds a local spectral summary. Let

$$X_k(H_{l,t}) = \{w_i - C_k(H_{l,t}) : i \in S_k(H_{l,t})\}$$

be the centered selected-embedding point cloud. We compute a low-rank spectrum of its covariance and denote the normalized leading eigenvalue vector by

$$\sigma_k(H_{l,t}) = (\sigma_1, \dots, \sigma_m).$$

The center+spectrum signature is

$$Z_k^{spec}(H_{l,t}) = (C_k(H_{l,t}), \sigma_k(H_{l,t})).$$

These three signatures form a small measurement family. They are deliberately derived from token identity rather than logit magnitude. The center tests whether the selected token set has a meaningful average location in W . The spread tests whether local selected-set dispersion matters. The spectrum tests whether internal shape of the token set adds further structure.

4.2 Proxy Topology-Preservation Metrics

For each layer l , each TopK size k , and each signature type, we compare the resulting TopK-derived representation with the original hidden-state representation.

Distance Spearman

Distance Spearman correlation compares pairwise distance orderings. If two hidden states are close in hidden space, their TopK-derived signatures should also be close. This gives a global test of distance-rank preservation.

KNN Overlap

KNN overlap compares local neighborhood structure. For each sample, we compute its nearest neighbors in hidden-state space and in TopK-signature space, then measure the overlap between the two neighbor sets.

CKA

Linear CKA measures global representational alignment between the hidden states and the TopK-derived signatures.

Composite Topology Score

we define a simple composite topology-preservation score:

$$\text{Topo} = \frac{1}{2} \rho_{\text{Spearman}} + \frac{1}{2} \text{KNNOverlap}_{10}.$$

This score is not meant to replace persistent homology. It is a practical summary of two complementary properties: global distance-order preservation and local neighborhood preservation. These metrics are proxy-topological diagnostics rather than persistent-homology reconstructions.

In this first audit, the goal is not to fully reconstruct the topology of the hidden-state manifold. The goal is more limited: to test whether the identity-only TopK set retains enough hidden-structural information to be treated as a meaningful vocabulary-side geometric observable.

4.3 Random Baseline and LiftTopo

To test whether the observed topology preservation could be reproduced by arbitrary vocabulary subsets, we construct a random-token baseline. For each hidden state and each TopK size k , we sample a random token set of the same cardinality:

$$S_k^{\text{rand}}(H_{l,t}) = \{j_1, \dots, j_k\}, \quad j_m \sim \text{Uniform}(V).$$

Using the same signature function applied to the true TopK set, we construct the corresponding random signature:

$$Z_k^{\text{rand}}(H_{l,t}) = \phi(S_k^{\text{rand}}(H_{l,t})).$$

where $\phi(\cdot)$ denotes one of the three identity-only signature maps: center, center+spread, or center+spectrum.

We then compute the topology-preservation score for the random baseline in the same way as for the true TopK signature:

$$\text{RandTopo} = \text{Topo}(H_l, Z_k^{\text{rand}}(H_l)).$$

The lift over random is defined as

$$\text{LiftTopo} = \text{Topo}(H_l, Z_k(H_l)) - \text{RandTopo}.$$

Here $\text{Topo}(H_l, Z_k(H_l))$ measures how well the true identity-only TopK set signature preserves the hidden-state geometry, while RandTopo measures how much of the same score can be reproduced by a random vocabulary subset of identical size. For this reason, LiftTopo estimates the excess topology-preservation signal attributable to the actual TopK token identity set rather than to the mere act of selecting k vocabulary vectors.

This baseline is important because large token sets may preserve some geometry simply by averaging over many vocabulary embeddings. A high LiftTopo rules out the weakest artifact explanation: that arbitrary token subsets of the same size can reproduce the observed result.

We also use three additional control families. A sample-shuffled TopK baseline preserves the global distribution of identity-only TopK sets but breaks the pairing between each hidden state and its own selected token set. An input-token baseline constructs signatures from tokens that appear in the input prompt, testing whether lexical overlap alone can account for the signal. A frequency-, token-type-, and token-length-matched baseline samples replacement tokens from buckets matched to the true TopK tokens, testing whether coarse token-distribution properties explain the effect.

Still, this control is not sufficient to eliminate all possible confounds. In particular, a uniform random-token set does not match token frequency, token type, token length, or anisotropy in the vocabulary embedding / output projection space. We therefore add a stronger supplementary analysis with frequency-, type-, and length-matched token sets. Randomized or shuffled- W baselines remain future work.

4.4 Experimental Environment and Model Backbone

All experiments were conducted on a local open-weight language model. The model backbone used in this experiment is

Qwen2.5-1.5B-Instruct.

The model was loaded from a local checkpoint and evaluated in inference mode. Hidden states were extracted from every Transformer block using forward hooks. For each input text, we recorded the hidden state at the final token position of each layer:

$$H_l = H_{l, t_{\text{last}}}.$$

The vocabulary readout matrix was taken directly from the model’s language-model head:

$$W = W_{\text{lm_head}}.$$

For each layer l , we computed the intermediate vocabulary readout

$$P_l = H_l W^\top,$$

and then constructed the identity-only TopK set

$$S_k(H_l) = \text{TopK}_k(P_l).$$

The TopK sizes tested in this experiment were

$$k \in \{20, 50, 100, 200, 500, 1000, 2000, 5000\}.$$

For each k , we evaluated three identity-only TopK-derived signatures:

$$\text{center}, \quad \text{center+spread}, \quad \text{center+spectrum}.$$

The experiment used the model’s output vocabulary embeddings as the geometric substrate for the identity-only TopK set. Before geometric comparison, vocabulary vectors were normalized when computing cosine-based distances and similarities. Hidden states were mean-centered layerwise before comparison.

The random baseline was repeated multiple times for each layer and TopK size. In the implementation used for this experiment, the number of random repetitions was

$$N_{\text{random}} = 5.$$

The maximum input length was set to

$$\text{MAX_LEN} = 160.$$

The implementation used PyTorch and HuggingFace Transformers. The model was evaluated on GPU with half-precision weights where available:

$$\text{dtype} = \text{float16}.$$

A fixed random seed was used for reproducibility:

$$\text{seed} = 42.$$

The purpose of this setup was not to benchmark model capability, but to audit whether the identity-only TopK set

$$\text{TopK}_k(H_l W^\top)$$

retains hidden-state neighborhood geometry under an identity-only constraint. For this reason, all reported results should be interpreted as object-level evidence for the observability of identity-only TopK sets, not as a claim about Qwen-specific model performance.

4.5 Statistical Reporting and Uncertainty

For the camera-ready version, each reported summary table should include the number of evaluation texts, the number of baseline sampling repetitions, and uncertainty estimates. The main deterministic model-forward pass uses a fixed text set; uncertainty should therefore be estimated by bootstrap resampling over evaluation texts and by repeated sampling for random, shuffled, input-token, and matched-token baselines.

The recommended reporting format is:

$$\text{Topo} = \mu \pm \text{SE}, \quad \text{Lift} = \mu \pm \text{SE}, \quad 95\% \text{ CI}_{\text{boot}}.$$

In addition, all figure captions should state whether error bars represent bootstrap confidence intervals over examples or variance over baseline-sampling seeds. The current manuscript should not be interpreted as relying on cross-training-seed variance; the models are fixed checkpoints. The relevant stochasticity comes from token-set baselines and from finite-sample evaluation.

5. Results

5.0 Preliminary Center Geometry Check

Before the main identity-only topology-preservation analysis, we first tested a simpler hypothesis: whether the center of the selected TopK token embeddings already preserves hidden-state geometry. This preliminary analysis uses the TopK token set

$$S_k(H_l) = \text{TopK}_k(H_l W^\top)$$

and maps it back into the vocabulary geometry through its center

$$C_k(H_l) = \frac{1}{k} \sum_{i \in S_k(H_l)} w_i.$$

The result is a clean scaling pattern. As k increases, the correlation between the TopK center and the hidden-state structure increases strongly, while random-token centers remain near zero. For example, at layer L0 the Spearman correlation rises from 0.1816 at $k = 20$ to 0.7568 at $k = 5000$, whereas the corresponding random baseline remains approximately zero.

This preliminary result is important for two reasons. It shows that the selected token identities are not an arbitrary list of labels; once mapped back into W , their centroid already carries hidden-structural information. it motivates the large-k selected-set interpretation used throughout this paper. The strongest signal appears at large k , which suggests that the relevant object is not a short next-token candidate list, but a wide readout-side selected set.

The main identity-only topology-preservation analysis in the next sections strengthens this preliminary result by removing logit values, probabilities, and rank scores, and by comparing the resulting TopK-derived signatures against random, shuffled, and input-token controls.

5.1 TopK Size Scaling: Large-k Selected Sets Preserve Hidden-State Geometry

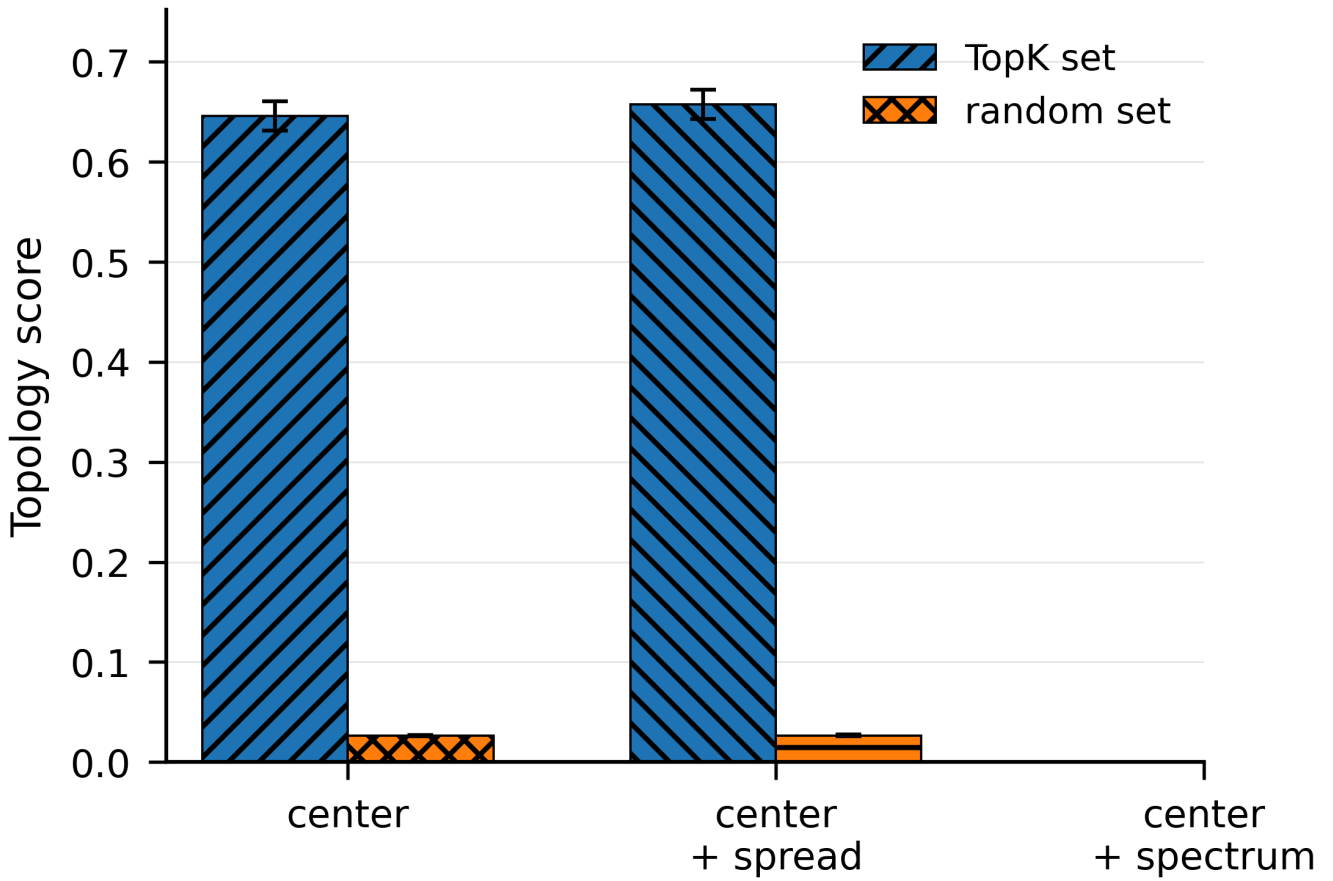


Figure 2. Identity-only TopK sets preserve hidden-state geometry above token-set baselines. This figure establishes the observable effect. It does not by itself attribute the effect to learned semantic structure in the vocabulary readout matrix. Error bars denote 95% confidence intervals across evaluated layer-condition rows for the fixed checkpoint; they do not represent cross-training-seed variance.

At $k = 5000$, the summary is:

Mode	TopK	MeanSp	MeanKNN	MeanCKA	MeanTopo	RandTopo	LiftTopo
center	5000	0.6513	0.6405	0.8722	0.6459	0.0266	0.6194
center+spread	5000	0.6595	0.6554	0.7875	0.6574	0.0266	0.6308
center+spectrum	5000	0.5635	0.5897	0.5820	0.5766	0.0265	0.5501

The strongest average signature is center+spread. At $k = 5000$, it reaches $\text{MeanTopo} = 0.6574$ and $\text{LiftTopo} = 0.6308$, while the random baseline remains near 0.0266.

One result is that the identity-only TopK set signatures are not close to the random-token baseline. Across all three identity-only signatures, the true TopK sets achieve substantially higher topology-preservation scores than random token sets of the same cardinality. This rules out the weakest artifact explanation: the effect is not reproduced merely by selecting k arbitrary vocabulary vectors.

The scaling with k is more revealing. The score grows as the selected embedding set widens. If the signal lived only in a few output candidates, small k should have been enough. It was not. The strongest preservation appears at the largest tested size, especially $k = 5000$.

A cautious reading is that $\text{TopK}_k(H_l W^\top)$ should not be understood primarily as a short next-token candidate list. In the regime where topology preservation is strongest, it behaves more like a large-k vocabulary-induced selected set: a finite subset of many related tokens that jointly preserves information about the hidden-state geometry.

Put differently, the experiment supports the following interpretation:

$$\text{TopK}_k(H_l W^\top) \neq \text{a small output-candidate list only.}$$

Rather, for sufficiently large k ,

$$\text{TopK}_k(H_l W^\top) \approx \text{a large-k readout set induced by } H_l.$$

This point is important for the rest of the paper. The object being studied is not merely the model's most likely next tokens, but a wider vocabulary region selected by the hidden state through the projection matrix W .

5.2 Layerwise Persistence: Identity-Only TopK Sets Are Not Only Final-Layer Artifacts

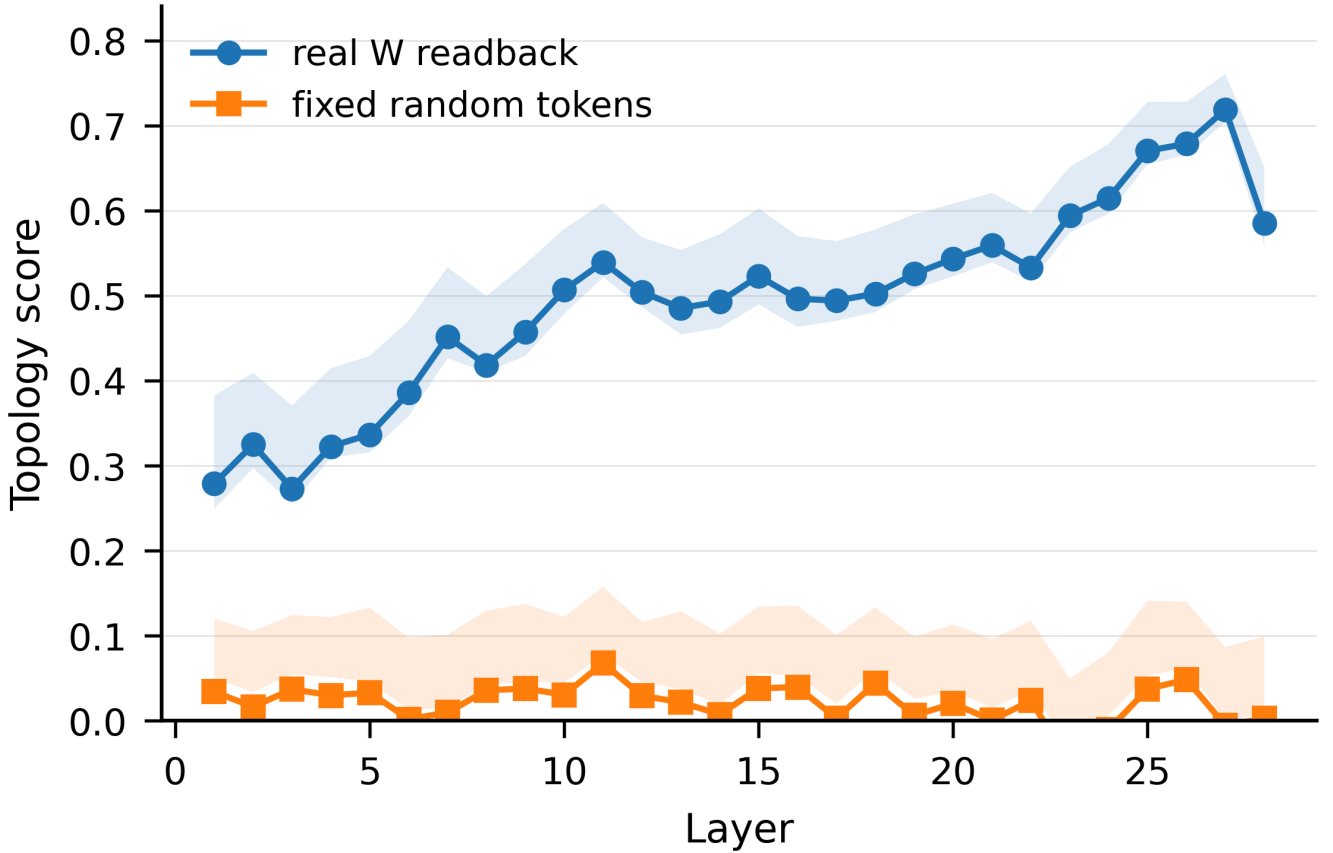


Figure 3. The readout-side observable remains above baseline across representative layers. Shaded bands denote bootstrap confidence intervals from the corrected W -disruption audit. Under the PSG interpretation, this is layerwise persistence of the readback observable, not proof of semantic row-identity structure preservation.

Figure 3 gives a second important result. The topology-preservation signal is not confined to the final layer. Instead, a stable signal is observable across the entire network depth.

Across all measured layers, the best identity-only signature remains well above the random baseline. The strongest layerwise result occurs at

$$L0, \quad k = 5000, \quad \text{mode} = \text{center+spread},$$

with

$$\begin{aligned} \text{Topo} &= 0.7471, \quad \rho_{\text{Spearman}} = 0.7753, \quad \text{KNNOverlap}_{10} = 0.7189, \\ \text{CKA} &= 0.8899, \quad \text{LiftTopo} = 0.7193. \end{aligned}$$

More importantly, this is not an isolated peak. The layerwise best profile shows that the same large- k selected-set signature remains strong throughout the network. For example, in the middle-to-late part of the network the signal is still substantial:

$$L20 : \quad \text{Topo} = 0.7105, \quad \text{LiftTopo} = 0.6849,$$

$$L21 : \quad \text{Topo} = 0.6948, \quad \text{LiftTopo} = 0.6665,$$

$$L22 : \quad \text{Topo} = 0.6934, \quad \text{LiftTopo} = 0.6646.$$

Even near the final layer, the signal does not collapse into the random baseline:

$$L27 : \quad \text{Topo} = 0.5945, \quad \text{LiftTopo} = 0.5717.$$

This layerwise persistence changes the interpretation of the object. If the topology-preservation signal appeared only at the final layer, a simple explanation would be available: the model might merely be arranging the vocabulary distribution for next-token generation. In that case, $\text{TopK}_k(H_l W^\top)$ would be closer to a final decoding artifact than an internal observable.

The data do not support that narrow interpretation. Since topology-preservation signals are visible across the full network depth, the identity-only TopK set appears to track structure already present in intermediate hidden states. This does not prove that identity-only TopK sets causally drive inference, but it does establish them as quantifiable internal observables rather than merely final-layer output lists.

So, Figure 3 supports the following conservative conclusion:

$\text{TopK}_k(H_l W^\top)$ retains hidden-structural information across layers, not only at the output boundary.

5.2.1 Why Does the Strongest Layer Occur at L0?

The strongest layerwise score occurs at L0. This is an important boundary condition rather than a result to hide. L0 is closest to the embedding interface. So part of the signal may reflect early embedding geometry and the direct coupling between the hidden state and the vocabulary readout. This makes the L0 peak compatible with a partial embedding-geometry explanation.

Still, the object-establishment claim does not depend on L0 alone. The same identity-only signal remains far above baseline in later layers, including the middle-to-late part of the network and the final layer. For example, L20–L22 remain around $\text{Topo} = 0.69$ – 0.71 , and L27 remains at $\text{Topo} = 0.5945$, with large lift over random. So, the strongest score may be shallow, but the effect is not shallow-only.

The conservative interpretation is therefore:

L0 shows the strongest embedding-coupled signal, while later layers show that the observable persists beyond the embedding boundary.

This distinction is important for the paper's scope. The present result establishes a vocabulary-level observable; it does not yet prove that this observable is a causal mechanism of inference.

5.3 Signature Ablation: Center, Spread, and Spectrum

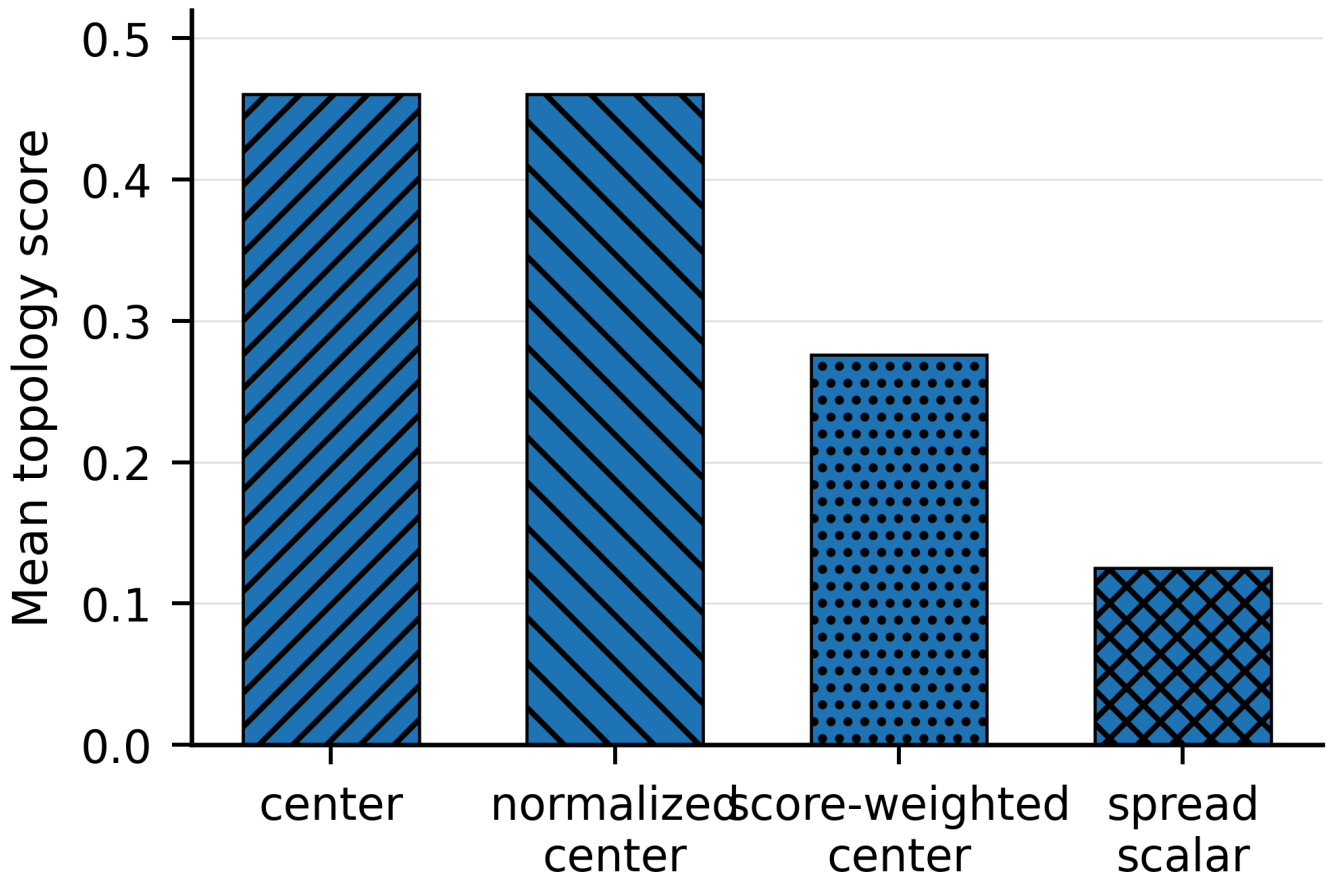


Figure 4. Centroid aggregation is the dominant contributor in the model-pipeline decomposition audit. Spread-like statistics are much weaker, which supports the conditional-mean readback interpretation. Bars show aggregate summary values.

Figure 4 summarizes the aggregation audit after the PSG realignment. The important comparison is no longer whether center+spread slightly improves on center in the original token-set signature table. The core mechanism is that the centroid itself carries the readback signal. A cautious reading is that the selected token identities are not an arbitrary collection of labels. Once mapped back into the readout geometry, their centroid $C_k(H_l)$ already retains substantial information about the hidden state that selected them.

Adding the spread term can improve the result in some settings, but the supplementary analysis shows that it does not universally dominate the center-only signature. This is still informative: a large-k readout set may contain useful volume information, but the selected-set centroid appears to carry the dominant low-order signal. Two identity-only TopK sets may have similar centers but different internal semantic dispersion profiles. The dispersion term $r_k(H_l)$ captures part of this difference. When center+spread improves over center alone, this suggests that selected-set dispersion is a meaningful component of the TopK geometry; when it does not, the result indicates that the centroid already captures most of the measurable topology-preservation signal.

Still, adding the covariance-spectrum signature does not further improve performance. This is a useful negative result. It shows that feature stacking does not automatically produce better topology preservation. A richer descriptor can add noise or unstable variation rather than useful structure.

A reasonable interpretation is that the topology-preservation signal is mainly carried by two low-order properties of the selected embedding set:

$$C_k(H_l)$$

and

$$r_k(H_l).$$

That is, the relevant information lies in where the selected embedding set is located in W , and how broadly it spreads around that location. By contrast, the higher-order covariance spectrum of the selected token set does not provide a clear additional gain in this audit.

A weaker reading is a cautious conclusion. For preserving hidden-state topology, the internal high-order association structure of the token set, such as logical or relational organization among selected tokens, is not yet shown to be necessary. The current evidence instead points to a lower-order geometric summary:

$$\boxed{\text{TopK topology preservation} \approx \text{selected-set centroid} + \text{semantic volume}.}$$

This does not mean that higher-order structure is absent. It only means that, under the present experiment measurements, the center explains most of the measurable topology-preservation signal, spread can provide a useful secondary descriptor, and the covariance-spectrum augmentation does not add a robust benefit.

5.4 Supplementary Control Analysis: Random, Shuffled, and Input-Token Baselines

To further test whether the topology-preservation signal could be explained by trivial vocabulary effects, we conducted a supplementary control audit with three baselines: random token sets, sample-shuffled TopK sets, and input-token baselines.

The purpose of this audit is slightly different from the main TopK scaling result. The main result asks whether identity-only TopK set signatures preserve hidden-state topology. The supplementary control audit asks what weaker explanations can be ruled out.

At $k = 5000$, the center signature achieves

$$\text{MeanTopo} = 0.5641.$$

The corresponding baselines are much lower:

$$\text{MeanRandTopo} = 0.1437,$$

$$\text{MeanShuffleTopo} = 0.0365,$$

and

$$\text{MeanInputTokenTopo} = 0.2637.$$

This yields the following excess scores:

$$\text{LiftRandom} = 0.4204,$$

$$\text{LiftShuffled} = 0.5276,$$

and

$$\text{LiftInputToken} = 0.3004.$$

These results strengthen the interpretation of the identity-only TopK set in three ways.

the signal is not reproduced by arbitrary token subsets of the same cardinality. This rules out the weakest explanation that large token sets preserve hidden geometry merely because they average over many vocabulary vectors.

the signal is not reproduced by sample-shuffled TopK sets. This is important because a shuffled TopK baseline preserves some global vocabulary-level properties of TopK selection while breaking the correspondence between a specific hidden state and its selected token set. The strong drop under shuffling suggests that the topology-preservation signal is sample-specific rather than merely a consequence of vocabulary-wide token bias.

the signal remains well above the input-token baseline. This addresses a different concern: perhaps the TopK set only works because it overlaps with lexical items already present in the prompt. The input-token baseline is substantially lower than the true TopK signal, suggesting that identity-only TopK sets contain more hidden-structural information than prompt lexical overlap alone.

The supplementary analysis therefore supports the following stronger interpretation:

identity-only TopK sets preserve hidden topology beyond random selection, shuffled vocabulary bias, and input-token lexical overlap.

A small caveat is also visible. In this supplementary analysis, center and center+spread are nearly tied. At $k = 5000$, the center signature reaches $\text{MeanTopo} = 0.5641$, while center+spread reaches $\text{MeanTopo} = 0.5639$. A cautious reading is that the selected-set centroid is the dominant low-order signal, while spread should be treated as a useful but not universally dominant augmentation.

5.5 Cross-Model Mini Replication of the Identity-Only Protocol

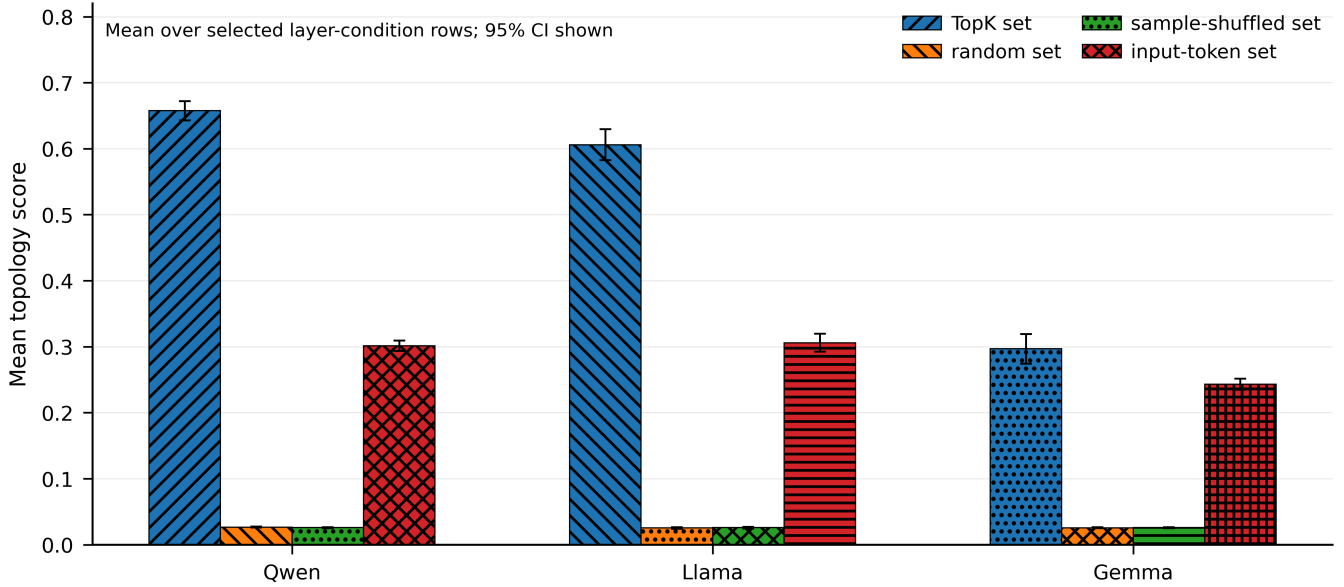


Figure 5. Cross-model observable replication under PSG-aware interpretation. Real TopK signatures remain above token-set baselines in Qwen, Llama, and Gemma, but this figure supports cross-model observability rather than semantic row-identity causality. Error bars denote 95% confidence intervals across evaluated layers for the selected TopK/mode condition within each fixed checkpoint; they do not represent training-seed replicates.

A major weakness of the earlier draft was that the primary identity-only topology-preservation audit was conducted only on Qwen. To address this, we ran an exact cross-model mini replication using the same identity-only protocol on Qwen, Llama, and Gemma. The replication uses the same aligned evaluation text set, the same identity-only signature family, and the same control families: random token sets, sample-shuffled TopK sets, and input-token baselines.

The best result for each model is:

Model	Best layer	TopK	Mode	Topo	Random	Shuffled	Input-token	Lift vs random	Lift vs shuffled	Lift vs input
Qwen	L0	5000	center+spread	0.747	0.028	0.027	0.362	0.719	0.720	0.385
Llama	L13	5000	center+spread	0.695	0.030	0.026	0.291	0.665	0.669	0.404
Gemma	L19	1000	center+spread	0.378	0.029	0.025	0.228	0.349	0.353	0.150

This result directly addresses the concern that the identity-only topology-preservation effect might be Qwen-specific. Qwen reproduces the original single-model result almost exactly. Llama shows a strong replication, with a best Topo = 0.695 and a large lift over all baselines. Gemma shows a weaker absolute score, but the effect remains positive and consistently above random, shuffled, and input-token controls.

The correct interpretation is therefore not that all models share the same absolute topology-preservation scale. They do not. The more conservative and more important result is that the direction of the effect replicates across model families:

$$\text{Topo}(\text{TopK}_{\text{real}}) > \text{Topo}(\text{random}), \quad \text{Topo}(\text{shuffled}), \quad \text{Topo}(\text{input token})$$

for Qwen, Llama, and Gemma.

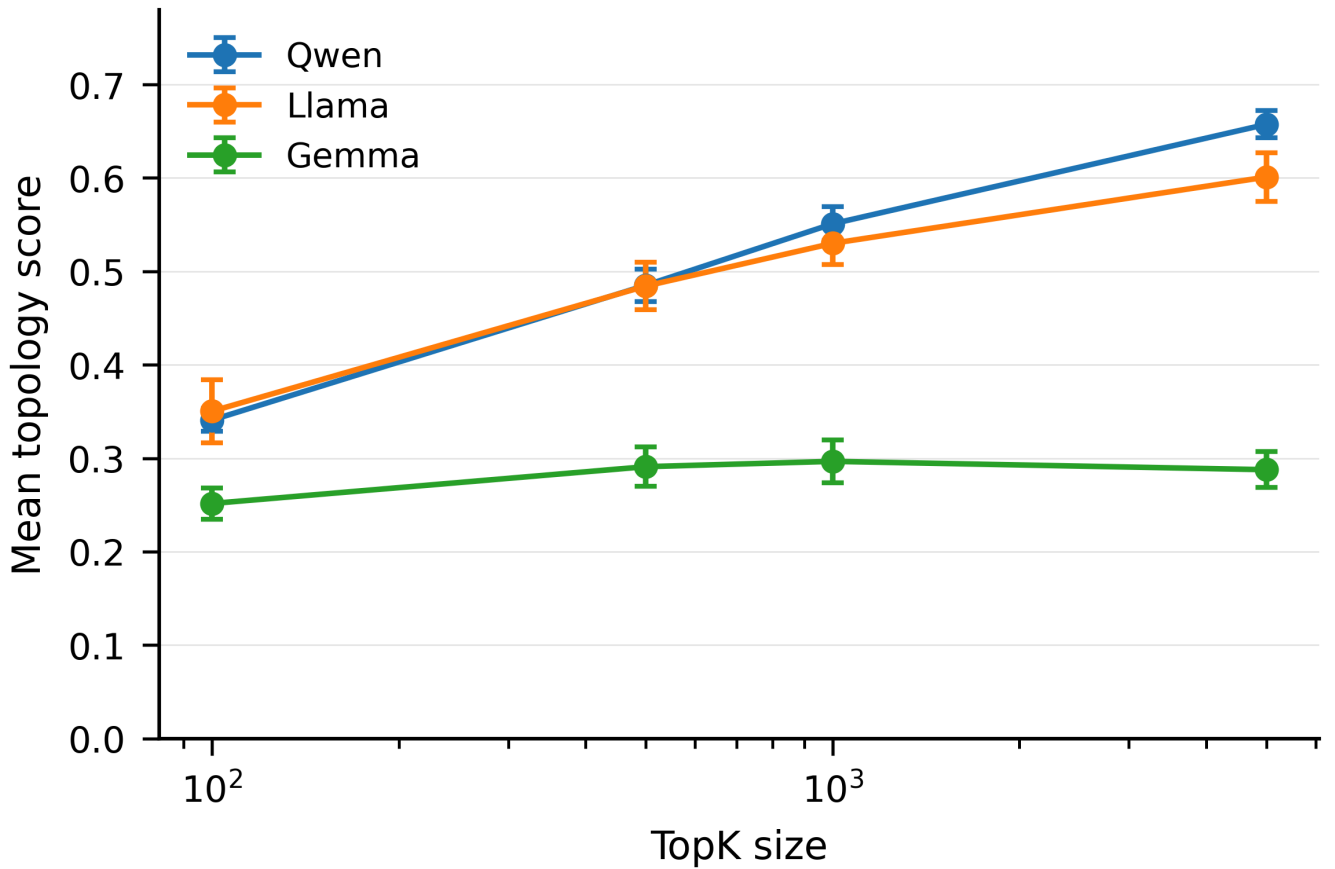


Figure 6. TopK size scaling across model families. Error bars denote 95% confidence intervals across evaluated layers for the center+spread signature within each fixed checkpoint. The large- k selected-set effect is reproducible, but under the conditional-mean interpretation this scaling is expected to reflect projection-selection readback strength as well as model-specific readout distribution.

Figure 6 shows that large selected sets remain important across models. Qwen and Llama both reach their strongest scores at $k = 5000$, while Gemma peaks at a smaller k in the layerwise best table. This model-specificity is consistent with the fact that the observable is mediated by the model’s own readout distribution, layer organization, and output embedding structure. What matters for this paper is not identical numeric scale, but reproducible separation from token-set baselines under a PSG-aware interpretation.

So, the cross-model result upgrades the main empirical claim from a single-model observation to a model-family-level phenomenon:

Identity-only identity-only TopK sets preserve hidden topology across multiple model families, with model-specific strength.

5.6 Frequency-, Token-Type-, and Token-Length-Matched Control

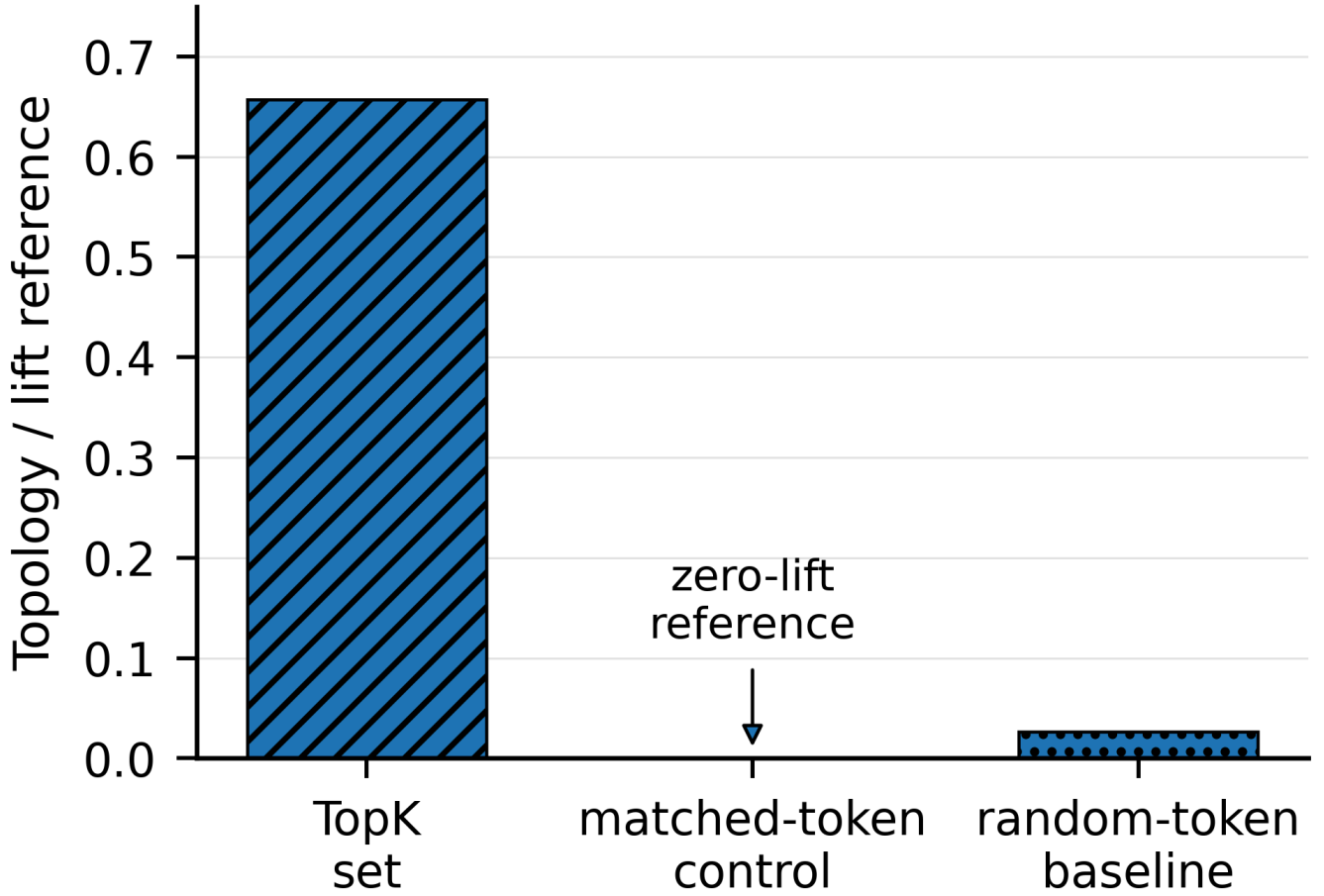


Figure 7. Token-statistic controls are early artifact checks. They help rule out simple token-frequency, token-type, and token-length explanations, but they do not establish semantic row-identity causality once the stronger PSG null is introduced.

The second major added control addresses token-distribution artifacts. A uniform random-token baseline does not match the true TopK set’s token frequency, token type, or token length distribution. For this reason, a positive result against uniform random token sets alone could still be explained by systematic properties of the selected tokens rather than by the state-specific selected vocabulary set.

To test this, we constructed matched token-set controls. For each true TopK set, replacement tokens were sampled from buckets that match token frequency, token type, and token length. This produces a stronger baseline than uniform random token selection:

$$S_k^{matched}(H_l) \sim \text{SampleByBucket}(\text{freq}(i), \text{type}(i), \text{len}(i)) .$$

The matched-control result is consistent across the tested grid. Across all 224 layer / TopK / signature combinations tested in Qwen, the real TopK signature remains above the matched baseline:

$$224/224 \text{ combinations have } \text{Lift}_{matched} > 0.$$

At the summary level:

TopK	Mode	Real Topo	Matched Topo	Lift vs matched
100	center	0.3309	0.0599	0.2710
100	center+spread	0.3409	0.0643	0.2766
500	center	0.4664	0.1265	0.3398
500	center+spread	0.4851	0.1324	0.3527
1000	center	0.5312	0.1586	0.3726
1000	center+spread	0.5509	0.1633	0.3877
5000	center	0.6459	0.2017	0.4442
5000	center+spread	0.6574	0.2025	0.4549

The residual lift trend is visualized in Appendix D.

The matched baseline explains some vocabulary-space structure: its topology score is higher than the uniform random and shuffled baselines. This is consistent with the fact that frequency, token type, and token length can carry geometric information in the vocabulary space. Still, the matched baseline remains far below the real TopK signal.

This result rules out a stronger artifact explanation than the random baseline did. The identity-only TopK topology-preservation effect cannot be explained solely by the marginal distribution of token frequency, token type, or token length. The state-specific membership of the real identity-only TopK set remains essential.

The strongest matched-control result occurs at

$$L0, \quad k = 5000, \quad \text{mode} = \text{center+spread},$$

where

$$\text{Topo}_{\text{real}} = 0.7471, \quad \text{Topo}_{\text{matched}} = 0.2716, \quad \text{Lift}_{\text{matched}} = 0.4755.$$

At the level of lift, the strongest point is

$$L2, \quad k = 5000, \quad \text{mode} = \text{center},$$

with

$$\text{Lift}_{\text{matched}} = 0.5646.$$

The conclusion is:

Frequency-, type-, and length-matched token-set artifacts explain only a minority of the topology-preservation signal.

The majority of the signal remains tied to the actual identity-only TopK set selected by the hidden state.

5.7 Corrected $\bar{W}$ -Disruption Audit: The Semantic- $\bar{W}$ Reading Fails

The reviewer-risk control changes the interpretation of the paper. Earlier baselines tested whether the signal could be explained by random token sets, shuffled sample-token pairings, input-token reuse, or coarse token statistics. Those controls narrowed the artifact space, but they left the readout geometry untouched.

The corrected $\bar{W}$ -disruption audit targets that gap. We reran the analysis on Qwen with 196 texts, last-non-punctuation pooling, layer 0 excluded, and 200 bootstrap replicates. Same-as-readout variants are treated only as sanity checks. The primary comparisons use disrupted readouts but realize the selected token sets in the original vocabulary geometry.

For center signatures at $k = 5000$, the main comparison is:

Control	Mean Topo	Mean Δ vs real	Mean 95% CI low	Mean 95% CI high
real $\bar{W}$	0.493	+0.000	0.471	0.566
row-permuted readout + original $\bar{W}$	0.619	+0.125	0.609	0.671
random-Gaussian readout + original $\bar{W}$	0.628	+0.135	0.620	0.678
orthogonal-rotated readout + original $\bar{W}$	0.630	+0.137	0.607	0.688
hidden orthogonal rotation + real $\bar{W}$	0.683	+0.189	0.661	0.732
fixed random tokens	0.021	-0.472	0.035	0.115
matched centroid-norm random tokens	0.027	-0.467	0.040	0.121

The pattern is not subtle. Fixed random-token baselines remain near zero, so ordinary random vocabulary geometry is not enough. Yet the disrupted readout controls are not lower than real $\bar{W}$. They are higher. The same broad pattern appears for center+spread:

Control	Mean Topo	Mean Δ vs real	Mean 95% CI low	Mean 95% CI high
real $\bar{W}$	0.362	+0.000	0.357	0.437
row-permuted readout + original $\bar{W}$	0.546	+0.184	0.537	0.607
random-Gaussian readout + original $\bar{W}$	0.523	+0.161	0.509	0.585
orthogonal-rotated readout + original $\bar{W}$	0.440	+0.079	0.428	0.513

Control	Mean Topo	Mean Δ vs real	Mean 95% CI low	Mean 95% CI high
hidden orthogonal rotation + real W	0.476	+0.115	0.457	0.548
fixed random tokens	0.022	-0.339	0.038	0.115
matched centroid-norm random tokens	0.028	-0.334	0.042	0.121

This forces a narrower conclusion. The observable preserves hidden-state geometry, but this preservation cannot be attributed specifically to learned semantic structure in W from the present evidence. The dominant remaining mechanism appears to be projection-selection geometry: a high-dimensional readout selects token vectors in a way that can encode pairwise geometry, even when semantic pairing is disrupted.

This result does not make the earlier positive results useless. It changes what they are evidence for. The paper now establishes a vocabulary-side geometric observable and a set of artifact boundaries. The semantic- W claim is left for a separate experiment, not asserted here.

5.8 Model-Pipeline Projection-Selection Decomposition Audit

The corrected W -disruption audit shows that real W is not uniquely responsible for TopK preservation. the model-pipeline decomposition audit tests this more directly by decomposing the projection-selection pipeline inside the model setting. It compares real W , row-permuted W , Gaussian score readouts, multiple realization matrices, TopK versus random-K selection, and several aggregation modes.

The result is decisive. TopK selection is far above random-K selection:

$$topk_mean = 0.6303, \quad randomk_mean = 0.0307.$$

Gaussian score readouts are not weaker than real W . In the reported Lite run, the mean topology-preservation score by score source is:

Score source	Mean topology-preservation score
gaussian_R	0.3830
row_permuted_W	0.3079
real_W	0.3006

The aggregation result is equally important. Center aggregation dominates spread-like summaries:

Aggregation	Mean topology-preservation score
center	0.4605
normalized_center	0.4605
score_weighted_center	0.2759

Aggregation	Mean topology-preservation score
spread_scalar	0.1250

Thus, the first mechanism-level conclusion is that TopK selection and centroid aggregation, not semantic row identity alone, are the dominant contributors to the preservation effect.

5.9 Pure Gaussian High-Dimensional Selection Null

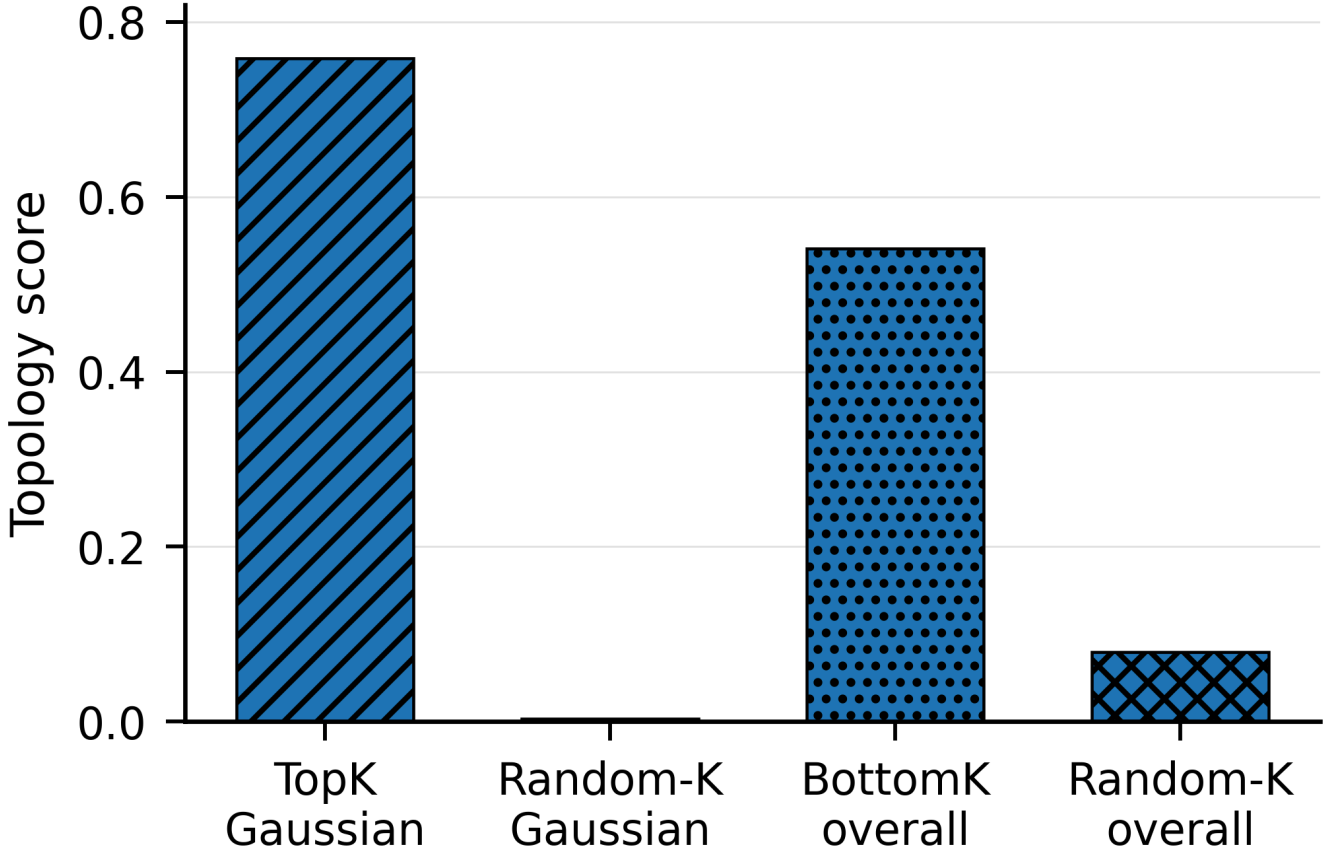


Figure 8. Pure Gaussian projection-selection null. With no model, tokenizer, vocabulary, or learned readout matrix, TopK centroid selection still preserves query geometry, while random-K carries little structure. BottomK is strong because it estimates the opposite query direction. Bars show aggregate summary values.

The pure Gaussian selection-null audit removes the model, tokenizer, vocabulary, and learned matrix entirely. Synthetic query vectors $H_i \in \mathbb{R}^d$ are projected onto random Gaussian readout vectors R_j , TopK indices are selected by $H_i R_j^\top$, and each selected set is represented by the centroid of its selected readout vectors:

$$Z_i = \frac{1}{k} \sum_{j \in \text{TopK}(H_i R^\top)} R_j.$$

The result is strong. In the pure Gaussian iid condition,

$$\text{gaussian_iid_topk_center_mean} = 0.7584, \quad \text{gaussian_iid_randomk_center_mean} = 0.0031.$$

Across conditions, TopK selection remains far above random-K:

$$\text{topk_mean} = 0.5381, \quad \text{randomk_mean} = 0.0797.$$

The result is not driven by set spread:

$$center_mean = 0.4809, \quad spread_scalar_mean = 0.0050.$$

BottomK is also strong:

$$bottomk_mean = 0.5410.$$

This is expected under the conditional-mean readback interpretation. TopK selects vectors with large positive projection on H_i , so its centroid estimates $+\alpha H_i$. BottomK selects vectors with large negative projection on H_i , so its centroid estimates $-\alpha H_i$. Since cosine geometry is preserved under global sign reversal, BottomK can preserve pairwise query geometry as well as TopK.

The mechanism can therefore be written as:

$$H_i R^\top \rightarrow ExtremeSelection_i \rightarrow mean(R_{Extreme_i}) \approx \pm \alpha H_i$$

This establishes projection-selection conditional-mean readback as a general high-dimensional readback mechanism.

5.10 Real- W Residual Audit

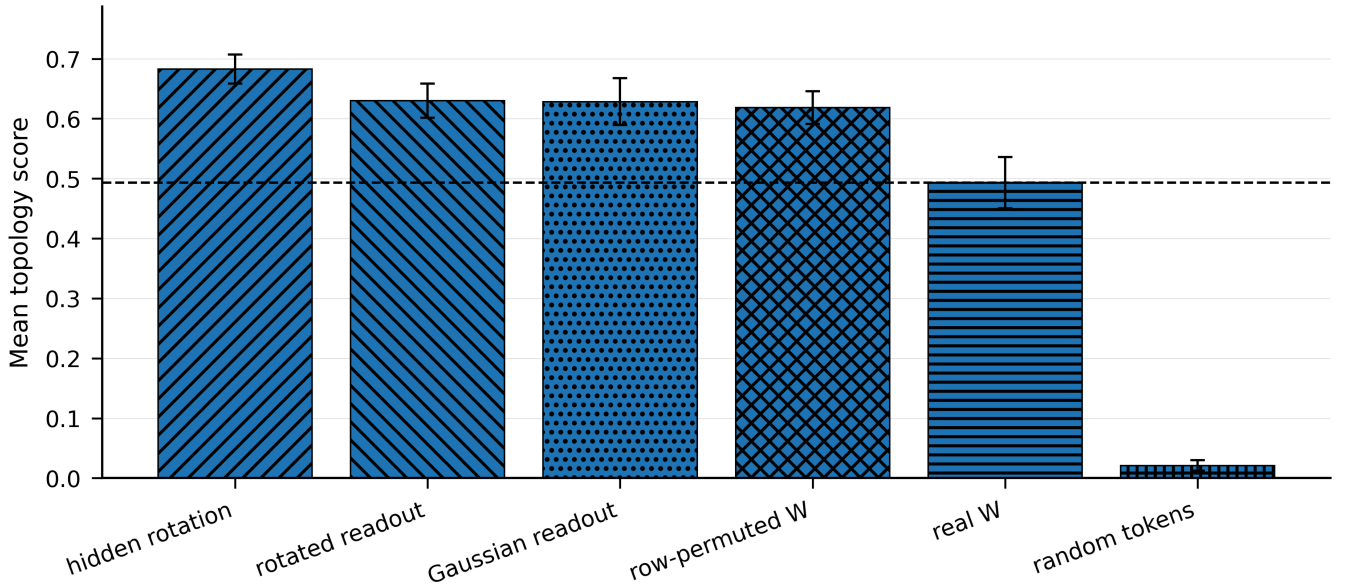


Figure 9. Real-readout residual audit with corrected W -disruption controls. Error bars denote 95% confidence intervals across evaluated layers for the fixed checkpoint. Randomized, permuted, and rotated readout controls match or exceed real W , while random-token controls remain low; this supports a negative semantic row-identity residual for topology preservation. The dashed reference line marks the real- W mean.

The real-readout residual audit asks whether real W contributes a positive residual beyond matched projection-selection nulls. The answer is negative. The best matched projection-selection null exceeds the best real- W condition:

$$real_best = 0.9331, \quad null_best = 0.9960, \quad real_best - null_best = -0.0629.$$

Row permutation does not materially affect the real- W readback score:

$$real_W_raw = 0.9286, \quad real_W_row_permuted = 0.9286.$$

This is incompatible with a strong semantic row-identity explanation of the topology-preservation effect. If token-row semantic organization were the main cause, row permutation should reduce preservation. It does not.

Gaussian matched nulls approach ceiling:

Condition	Score
gaussian_iid	0.9960
gaussian_norm_matched	0.9960
gaussian_cov_matched_diag	0.9928

Whitening reduces real- W performance:

$$real_W_whiten_row_norm = 0.8584 < real_W_raw = 0.9286.$$

Thus, the raw distributional geometry of W , including covariance or anisotropy, can modulate readback. But this modulation is not evidence for token-level semantic row-identity structure, because row permutation does not hurt and matched Gaussian nulls are stronger.

The final interpretation is:

$$ObservedTopKPreservation = projection - selectionconditional - meanreadback + Wdistributionalmodulation + nodetectedpositivesemanticrow - identityresidual$$

5.11 Conditional-Mean Readback Mechanism

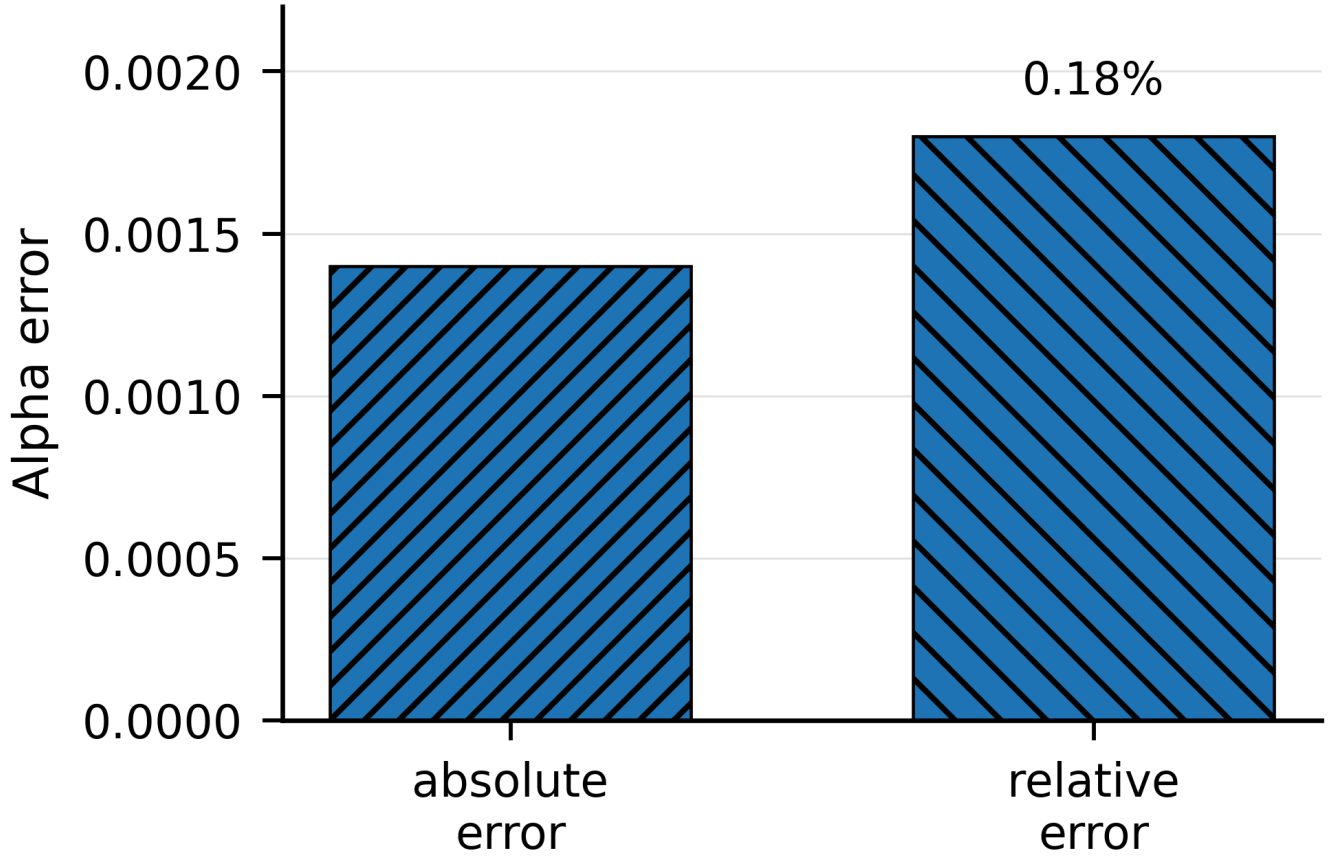


Figure 10. Conditional-mean readback audit. The empirical readback coefficient matches the conditional-mean prediction with very small error, promoting PSG from an empirical confound to a theorem-like mechanism. Bars show absolute and relative alpha-fit errors.

The conditional-mean readback audit tests whether the readback effect is explained by a conditional-mean law rather than by an empirical artifact of the simulation. For isotropic Gaussian candidates, let

$$s = h \cdot R, \quad s \sim \mathcal{N}(0, 1),$$

and let TopK selection correspond to the event

$$s \geq t, \quad t = \Phi^{-1}(1 - k/V).$$

Then the conditional mean of the selected projection is

$$\alpha_{theory} = \mathbb{E}[s \mid s \geq t] = \frac{\phi(t)}{1 - \Phi(t)}.$$

For sphere-normalized candidates, the same expression is scaled by $1/\sqrt{d}$:

$$\alpha_{theory} \approx \frac{1}{\sqrt{d}} \frac{\phi(t)}{1 - \Phi(t)}.$$

The empirical estimate matches this prediction closely. The reported mean errors for TopK are

$$topk_alpha_abs_err = 0.0014, \quad topk_alpha_rel_err = 0.0018.$$

Thus, the readback equation should be written as a noisy conditional-mean estimator:

$$Z_i = \frac{1}{k} \sum_{j \in \text{TopK}(H_i R^\top)} R_j = \alpha(d, V, k) H_i + \epsilon_i.$$

The residual is not zero. the conditional-mean readback audit reports a moderate residual ratio and high self-alignment:

$$\text{topk_residual_ratio} = 0.3525, \quad \text{topk_self_cos} = 0.9108.$$

The mechanism is therefore not a perfect reconstruction of H_i . It is a noisy estimator that is nevertheless strong enough to preserve pairwise geometry. The geometry result remains strong:

$$\text{topk_topo_spearman} = 0.8213, \quad \text{bottomk_topo_spearman} = 0.8216.$$

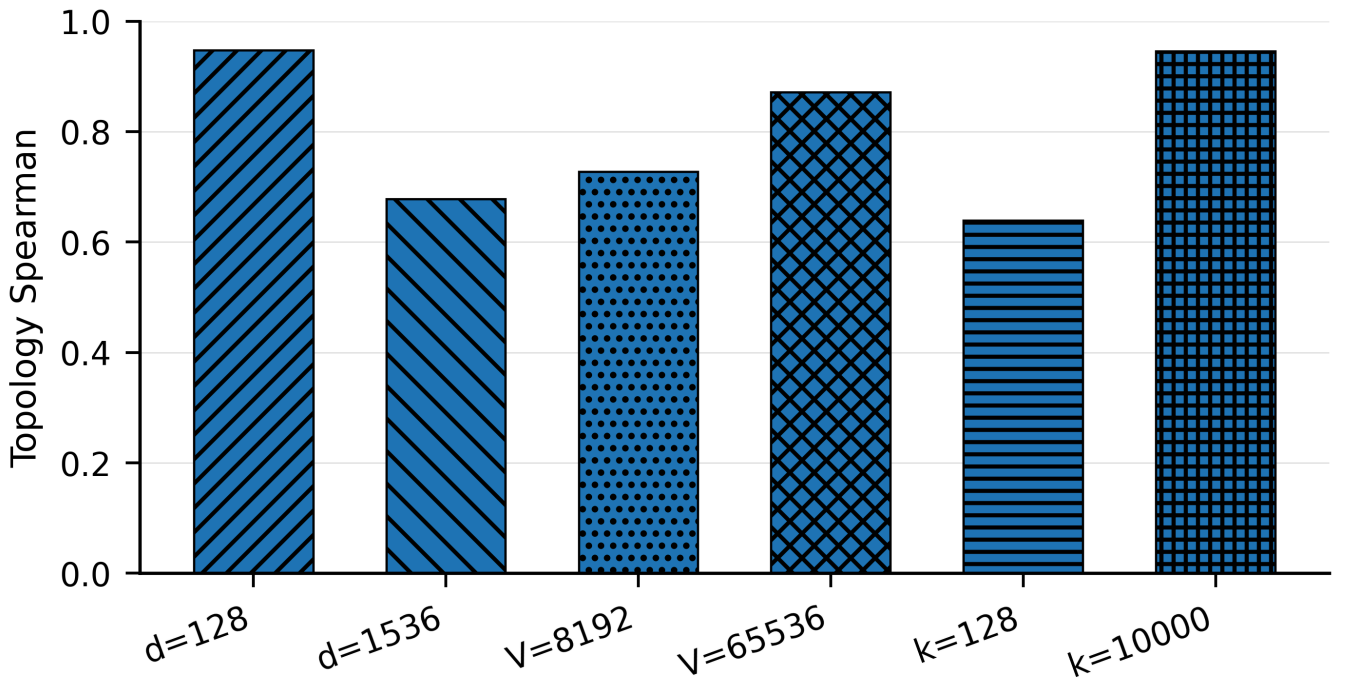


Figure 11. PSG readback scaling summary. Readback strength changes systematically with dimension, candidate-pool size, and TopK size. These trends support the conditional-mean estimator interpretation and are not specific to learned vocabulary semantics. Labels summarize representative sweep endpoints.

This section closes the mechanism: TopK preservation is explained by conditional selection on extreme projection, which shifts the selected centroid along the query direction. BottomK gives the opposite sign, and random-K lacks the conditioning event.

6. What Does Not Work? Failure Regimes and Boundary Conditions

The negative results are important because they prevent the object from being over-interpreted. The experiment does not show that any TopK set, any signature, or any model family gives the same preservation effect. Instead, the failures identify the regime in which the observable is meaningful.

6.1 Small-k Is Weak

Small selected sets are weak. When $k < 100$, topology-preservation scores drop substantially. This is the first boundary condition: the object should not be interpreted as a short next-token shortlist. The stronger signal appears only when TopK is broad enough to behave as a selected embedding set.

6.2 Spectrum Does Not Help

The center+spectrum signature does not improve the result. In several settings it is weaker than center or center+spread. This failure is useful because it rules out a simple feature-stacking explanation. More spectral information is not automatically better.

6.3 Gemma Is Weaker

The effect direction replicates across Qwen, Llama, and Gemma, but the magnitude is model-specific. Gemma is substantially weaker in absolute terms. The conservative interpretation is that identity-only TopK set is not a model-independent scalar effect. It is an observable whose strength depends on the geometry of the model family.

6.4 Large-k Selected-Set Boundary

The strongest setting is $k = 5000$. This should not be read as a universal optimum. It is better interpreted as evidence for a large-k selected-set regime: the TopK set must be large enough to sample a vocabulary-level selected readout set rather than a narrow decoding candidate list.

7. What the Experiment Rules Out

The corrected W-disruption result changes this section from a victory list into a boundary map. Several easy explanations are still ruled out. One harder explanation is not. The distinction is now central to the paper.

This experiment does not prove everything we would like to know about identity-only TopK sets. It does not prove causality, and it does not yet rule out all possible artifacts. However, it does rule out several weak but important alternative explanations, especially after the supplementary selected-set control audit.

7.1 The Topology-Preservation Signal Is Not Simply Carried by Logit Magnitudes

A first possible explanation is that the observed topology-preservation signal comes from the full logit vector itself. This would be a weak result. Since

$$P_l = H_l W^\top$$

is a linear readout of the hidden state, it is expected to retain part of the geometry of H_l . If the experiment only showed that full logits preserve hidden-state structure, the result would be difficult to interpret. It could simply reflect the fact that a high-dimensional linear projection still contains substantial information about its input.

This experiment removes this explanation by discarding logit magnitudes before constructing the tested signatures. The experiment does not use the values of $P_{l,i}$. Instead, it keeps only the identities of the selected tokens:

$$S_k(H_l) = \text{TopK}_k(H_l W^\top).$$

The center, center+spread, and center+spectrum signatures are all derived from this token identity set. For this reason, the measured topology-preservation signal cannot be attributed directly to the numerical size of the logits.

The result should be read carefully. It does not mean that logits are irrelevant to the model. Logits determine which tokens enter the TopK set. But once the set is selected, the topology-preservation measurement no longer uses the logit values themselves. The signal survives after the magnitude information has been removed.

7.2 The Signal Is Not Simply Carried by Probabilities or Rank Scores

A second possible explanation is that the topology signal comes from probabilities or fine-grained rank information. For example, one might suspect that the probability gap between the top tokens, or the detailed order of tokens inside the TopK set, is what preserves the hidden-state geometry.

This experiment also removes this explanation. The identity-only test does not use probability values, softmax weights, rank scores, logit gaps, or within-TopK ordering weights.

The tested object is not a weighted distribution over the selected tokens. It is the selected token set itself. In the strongest form, the experiment asks whether the set

$$\{i_1, \dots, i_k\}$$

contains structural information after all score-like quantities have been discarded.

This is why the result is stronger than a standard logit or probability-space analysis. If the topology-preservation signal depended mainly on probability mass or rank gaps, then removing those quantities should have destroyed most of the effect. Instead, the identity-only signatures still preserve substantial hidden-state topology. A cautious reading is that the membership of the identity-only TopK set carries structure, not merely the probability values assigned to its members.

Again, this does not imply that ranking is meaningless. The ranking process is how the set is produced. But the experiment shows that after the set has been selected, the identities of the included tokens are already informative.

7.3 The Signal Is Not Reproduced by Random Token Selection

A third possible explanation is that any sufficiently large subset of vocabulary embeddings would preserve some hidden-state geometry. This concern is especially important because the strongest results appear at large k , particularly $k = 5000$. Large token sets could, in principle, produce stable geometric summaries simply by averaging over many vocabulary vectors.

To test this, the experiment constructs random-token baselines. For each hidden state and each TopK size k , a random token set of the same cardinality is sampled:

$$S_k^{\text{rand}}(H_l) = \{j_1, \dots, j_k\}, \quad j_m \sim \text{Uniform}(V).$$

The same signature construction is then applied to the random set:

$$Z_k^{\text{rand}}(H_l) = \phi(S_k^{\text{rand}}(H_l)),$$

where ϕ denotes the center, center+spread, or center+spectrum signature map.

The random baseline topology score is

$$\text{RandTopo} = \text{Topo}(H_l, Z_k^{\text{rand}}(H_l)).$$

The excess signal above random is measured by

$$\text{LiftTopo} = \text{Topo}(H_l, Z_k(H_l)) - \text{RandTopo}.$$

The empirical result is that true identity-only TopK set signatures remain far above the random baseline. This rules out the weakest random-selection explanation: the observed topology preservation is not reproduced by arbitrary vocabulary subsets of the same size.

The supplementary analysis further adds two stricter controls. In the sample-shuffled TopK baseline, identity-only TopK sets are decoupled from the hidden states that produced them. The resulting topology score is much lower than the real TopK score, which suggests that the signal is not merely a vocabulary-wide selection bias. The hidden state and its own induced selected token set must remain paired.

The input-token baseline provides another control. It tests whether the topology signal can be explained simply by lexical overlap with the prompt. The real identity-only TopK sets remain substantially above this baseline as well. the topology-preservation signal is not merely a reflection of tokens already present in the input text.

However, this control has a clear boundary. Random token sets do not control for token frequency, tokenization structure, or anisotropy in the vocabulary embedding space. this experiment rules out arbitrary random selection, but it does not yet rule out all vocabulary-space artifacts. Stronger future controls should include tokenization-aware baselines, rank-shuffled TopK sets, and randomized, permuted, or rotated- W baselines.

7.4 The Signal Is Not Explained by Frequency-, Type-, and Length-Matched Token Sets

The frequency-, token-type-, and token-length-matched control rules out a stronger artifact explanation. Instead of sampling arbitrary vocabulary items, this baseline samples replacement tokens from buckets matched to the true TopK tokens. It therefore preserves coarse token-distribution properties that uniform random sets do not.

If the topology-preservation signal were mainly caused by high-frequency tokens, token type, or tokenizer length, the matched baseline should approach the true TopK score. It does not. The matched baseline is consistently higher than uniform random baselines, showing that these token statistics do explain some structure, but it remains far below the real TopK signal. Across all tested layer / TopK / mode combinations, the lift over the matched baseline remains positive.

This supports the following stronger claim:

$\text{TopK}_k(H_l W^\top)$ preserves hidden-state topology beyond frequency-, type-, and length-matched token-set artifacts.

This does not eliminate all vocabulary-space confounds. In particular, output-embedding anisotropy and deeper semantic-frequency structure remain possible contributors. But it substantially narrows the artifact space.

8. Remaining Confounds

The added controls strengthen the paper, but they do not close the artifact space. Token frequency, token type, and token length are no longer untested in the main Qwen setting. They are also not eliminated in every possible form.

8.1 Frequency and Tokenization Are Partially Controlled, Not Fully Eliminated

The frequency-, token-type-, and token-length-matched audit shows that the real TopK signal remains far above matched token-set controls. This is a strong defense against the claim that the effect is merely due to high-frequency tokens, tokenizer length, or coarse token category.

However, this does not fully eliminate every frequency- or tokenization-related explanation. The control matches bucketed statistics, not the entire distributional history of each token. More subtle effects may remain, including domain-specific token frequency, semantic-frequency interactions, capitalization patterns, subword morphology, or local token-cluster density in W .

the current conclusion should be:

Coarse frequency/type/length artifacts do not explain the effect.

It should not be overstated as:

All frequency and tokenization artifacts are impossible.

8.2 Projection-Selection Geometry Is Not a Remaining Confound, but the Main Mechanism

Earlier versions of this paper treated projection-selection geometry as a remaining confound. the model-pipeline, pure Gaussian, and real-readout residual audits change its status. The projection-selection conditional-mean readback mechanism is not a minor artifact to be controlled away after establishing a semantic claim. It is the dominant explanation of the topology-preservation effect.

The mechanism is simple. In high dimensions, extreme selection by $H_i R^\top$ chooses readout vectors aligned with the query direction. Averaging those selected vectors produces a centroid close to H_i . Selecting the opposite extreme produces a centroid close to $-H_i$. Random-K selection lacks this alignment and therefore carries little pairwise geometry.

This means topology preservation is not sufficient for semantic attribution. A TopK set can preserve hidden geometry because it is a projection-selection readback estimator, even if R is random Gaussian. Any future semantic claim must therefore be residualized against a matched projection-selection null.

8.3 Model-Specific Strength

The cross-model mini replication shows that the effect is not Qwen-only. However, it also shows model-specific strength. Qwen and Llama produce strong absolute Topo scores, while Gemma shows a weaker but still positive replication.

The implication is the claim should not be written as:

All models preserve topology equally well.

The supported claim is:

The effect direction replicates across model families, but its strength is model-specific.

This model-specificity is expected because vocabulary geometry, layer organization, tokenizer design, and output embedding structure differ across models.

8.4 Proxy-Topological Metrics and Persistent Homology Interface

The composite score Topo remains a practical proxy. It combines distance-rank preservation and KNN overlap, but it is not equivalent to full persistent homology. The result should therefore be read as evidence of hidden-geometry preservation in a proxy-topological sense, not as a complete topological reconstruction.

The next methodological step should be to combine identity-only TopK set signatures with persistent homology or graph-based topology diagnostics.

9. Scholarly Value of the Corrected Object

9.1 Why This Object Matters

The object matters for a different reason than the initial semantic reading suggested. The central contribution is not that TopK preservation proves learned semantic row-identity preservation. It does not. The contribution is that identity-only TopK sets expose a strong and general readout geometry that is easy to overinterpret if left unaudited.

This has direct value for mechanistic interpretability. Vocabulary-side observables are attractive because they appear closer to human-readable semantics than hidden vectors. But this paper shows that a vocabulary-side TopK observable can preserve hidden geometry even when the readout is random. The observable is real, but the semantic attribution is not automatic.

The corrected object is therefore useful in two ways. First, it provides a practical score-free observable of hidden-state geometry. Second, it supplies a necessary null model for future work: any claim that TopK reveals learned semantic structure in $\mathcal{W}$ must demonstrate residual structure beyond projection-selection conditional-mean readback.

9.2 Future Work: Layerwise Selected-Set Variation

The paper establishes observability rather than a theory of layerwise change. It shows that identity-only TopK set signatures preserve substantial hidden-state topology even when only token identities are retained. This is sufficient to justify future studies of how these selected sets vary across layers.

Such future work should treat the present result as a precondition, not as a dynamical theory. The current version does not model transition laws, causal mechanisms, or the internal process that produces final outputs. It only establishes that $\text{TopK}(H_l W^\top)$ is a non-arbitrary vocabulary-level observable whose layerwise behavior may be worth studying.

9.3 Future Work: Beyond Proxy Topology

This work uses distance Spearman correlation, KNN overlap, CKA, and a composite TopoScore as proxy-topological diagnostics. These metrics are useful for a first object-establishment audit, but they are not equivalent to persistent homology or a complete topological reconstruction.

The next useful step is to combine identity-only TopK set signatures with persistent homology, zigzag persistence, graph topology, and geometry-disrupted vocabulary controls. Such work would test whether the topology-preservation signal reported here survives stricter topological diagnostics and stronger interventions on the vocabulary geometry.

9.4 Supplementary Evidence Hierarchy

The revised supplement includes an exact cross-model mini replication of the identity-only topology-preservation protocol. This is the main cross-model support for the present paper because it tests the same object under the same identity-only constraint across Qwen, Llama, and Gemma.

The evidence hierarchy is as follows.

Evidence status	Evidence family	Role
Primary evidence	Identity-only TopK topology-preservation analysis	Main object-establishing result
Primary controls	Random-token, sample-shuffled, input-token, and frequency/type/length-matched baselines	Main artifact-defense evidence
Cross-model supplement	Exact identity-only mini replication on Qwen, Llama, and Gemma	Direct multi-model support for the same protocol
Secondary convergent evidence	Cone-inversion audit	Useful but secondary, because it mixes token identity with logit values and numerical inversion
Excluded from the Paper1 claim	Broader non-identical TopK protocols	Possible motivation for later work, but not required for the object-establishing claim

The revised conclusion is therefore stronger but still conservative: identity-only TopK set signatures preserve hidden-state topology across multiple model families, with model-specific effect strength.

10. Limitations

The main limitation is now part of the paper's central result. TopK token-set preservation should not be interpreted as direct evidence of learned vocabulary geometry. the pure Gaussian selection-null audit shows that pure Gaussian readouts can produce strong conditional-mean readback without any model or vocabulary. the real-readout residual audit further shows that matched projection-selection nulls can exceed real $\mathcal{W}$, and that row permutation does not materially reduce preservation. The present paper therefore establishes a readout observable and a readout mechanism, not a semantic mechanism of inference.

The second limitation is that the residual semantic contribution of W remains unresolved in a narrower sense. the real-readout residual audit is negative for topology-preservation residuals, but real W may still contribute semantic structure through observables not captured by centroid preservation. Future work should examine projection-selection geometry-orthogonal features: TopK set turnover, entry/exit asymmetry, rank-flow structure, semantic category enrichment, non-centroid boundary tokens, and condition-specific layerwise transitions.

The third limitation concerns metrics. Distance Spearman, KNN overlap, CKA, and the composite Topo score are proxy-topological diagnostics. They are useful for fast auditing, but they do not replace persistent homology or richer topological analysis.

The fourth limitation is model coverage. The main line uses Qwen as the primary backbone, with supplementary mini-replication on Llama and Gemma. The projection-selection null itself is model-free, but the language-model readout observations should still be replicated across more model families, sizes, tokenizers, and output-head tying conventions.

11. Conclusion

This paper began with a simple observation: identity-only TopK sets preserve hidden-state neighborhood geometry after logits, probabilities, rank scores, and within-set weights are removed. That observation remains true. But the interpretation changes substantially after the projection-selection geometry audits.

The preservation effect is not sufficient evidence for learned semantic row-identity causality. Corrected W -disruption and real- W residual audits show that matched Gaussian and permuted readout constructions can match or exceed real W , and row permutation does not materially reduce the effect. Pure Gaussian projection-selection controls then show why: extreme projection selection shifts the centroid of selected candidate vectors along the query direction. The conditional-mean readback audit verifies this as a conditional-mean mechanism.

The final claim is therefore narrower and stronger. Identity-only TopK sets are valid readout-side observables of hidden-state geometry. Their topology-preservation effect is primarily explained by projection-selection conditional-mean readback:

$$Z_i = \frac{1}{k} \sum_{j \in \text{TopK}(H_i R^\top)} R_j = \alpha(d, V, k) H_i + \epsilon_i.$$

Learned W may still carry semantic content and may modulate readback through distributional geometry. But topology preservation alone is not a semantic attribution test. Future work should therefore focus on projection-selection geometry-orthogonal observables: layerwise set turnover, entry/exit asymmetry, semantic category residuals, mechanism-label discrimination, and other structures that cannot be reduced to centroid readback.

Appendix A. Complementary Cone-Inversion Analysis

A separate TopK cone-inversion analysis provides complementary but secondary evidence for the role of token identity. Unlike the main identity-only audit, this experiment attempts to reconstruct hidden-state geometry from TopK-derived information using ridge-style inversion. It compares settings that use only TopK logit values with settings that use both logit values and token identities.

The main qualitative result is that adding token identity often improves hidden-geometry reconstruction compared with using logit values alone, especially in larger- k regimes. For example, at L0 and $k = 2000$, the values-only setting gives

$$\text{RelMSE} = 1.1303, \quad \text{Cos} = 0.2803, \quad R^2 = -0.1504, \quad \text{CKA} = 0.5039, \quad \text{Topo} = 0.2376.$$

The corresponding values+ids setting gives

$$\text{RelMSE} = 0.7058, \quad \text{Cos} = 0.3823, \quad R^2 = 0.2816, \quad \text{CKA} = 0.5504, \quad \text{Topo} = 0.3448.$$

This supports a weaker but useful claim: token identity is not a disposable byproduct of logit magnitude. It carries additional geometric information that can improve reconstruction of hidden-state structure. However, this cone-inversion analysis is not used as primary evidence in the paper. It mixes logit values, token identities, and numerical inversion, and some regimes show ill-conditioned fitting. For that reason, it is best interpreted as supplementary evidence that converges with the main result, rather than as the central proof of topology preservation.

The primary evidence remains the identity-only topology audit, where logit values, probabilities, and rank scores are removed before constructing the TopK-derived signatures.

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Appendix B. Evidence Chain Summary

The paper is organized around an object-establishing claim:

$$S_k(H_l) = \text{TopK}_k(H_l W^\top), \quad \mathcal{E}_k(H_l) = \{w_i : i \in S_k(H_l)\} \text{ define a meaningful vocabulary-side geometric observable of hidden-state structure.}$$

The current evidence chain is as follows.

Evidence	Role in Paper 1	Main implication
TopK Token Center Geometry Check	Preliminary center-geometry evidence	TopK token centers preserve hidden-state geometry, and the signal strengthens with larger k .
Identity-Only TopK Topology Preservation Analysis	Primary identity-only topology audit	TopK token identity alone preserves hidden-state topology after removing logit values, probabilities, and rank scores.
Selected-Set Control Analysis with Random, Shuffled, and Input-Token Baselines	Stronger artifact controls	The signal remains above random-token, sample-shuffled, and input-token baselines.
Exact Cross-Model Mini Replication	Direct cross-model replication	Qwen and Llama show strong replication; Gemma shows weaker but positive replication.

Evidence	Role in Paper 1	Main implication
Frequency/Type/Length-Matched Baseline	Stronger artifact defense	The signal remains above token sets matched for frequency, token type, and token length.
Top-k Cone Inversion Curve	Supplementary cone-inversion evidence	Token identity improves hidden-geometry reconstruction beyond logit values alone in many regimes, but the audit is numerically less stable and remains supplementary.

These results have different evidential status. The Identity-Only TopK Topology Preservation Analysis remains the main object-establishing experiment. The Selected-Set Control Analysis, the cross-model mini replication and the matched-token control provide the main defensive evidence against weak artifact explanations and single-model overfitting. The TopK center audit explains why the selected-set centroid construction is meaningful and why large k should be interpreted as a large-k readout set. The cone-inversion analysis provides convergent but secondary support that token identity contributes geometric information beyond values, but it is not a direct identity-only topology-preservation test.

The conservative conclusion is therefore:

Broad identity-only TopK sets preserve hidden-state structure beyond several weak artifact explanations.

This paper establishes the middle object: a vocabulary-induced vocabulary-side geometric observable that bridges hidden-state topology and vocabulary-level geometry.

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Appendix C. Exact Cross-Model Mini Replication

This appendix reports the direct cross-model replication of the identity-only TopK topology-preservation protocol.

C.1 Configuration

Field	Value
Models	Qwen2.5-1.5B-Instruct, Llama-3.2-1B-Instruct, Gemma-2-2B-it
Text set	aligned evaluation text set
TopK	100, 500, 1000, 5000
Signatures	center, center+spread
Baselines	random, shuffled, input-token
Random repetitions	5
Max length	160

C.2 Best per Model

Model	Best layer	TopK	Mode	Topo	Random	Shuffled	Input-token	Lift vs random	Lift vs shuffled	Lift vs input
Qwen	L0	5000	center+spread	0.7471	0.0277	0.0274	0.3619	0.7193	0.7196	0.3852
Llama	L13	5000	center+spread	0.6948	0.0298	0.0262	0.2908	0.6650	0.6686	0.4040
Gemma	L19	1000	center+spread	0.3778	0.0290	0.0251	0.2282	0.3488	0.3527	0.1496

C.3 Interpretation

The replication supports the following claim:

The identity-only TopK topology-preservation effect is not Qwen-specific.

However, the strength is model-specific. Qwen and Llama show strong absolute topology-preservation scores; Gemma shows a weaker but still positive replication. This should be reported as cross-model positive replication, not as identical cross-model behavior.

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Appendix D. Frequency-, Token-Type-, and Token-Length-Matched Baseline

This appendix reports the matched-token control audit.

D.1 Configuration

Field	Value
Model	Qwen2.5-1.5B-Instruct
Text set	aligned evaluation text set
TopK	100, 500, 1000, 5000
Signatures	center, center+spread
Baselines	random, shuffled, frequency/type/length matched
Matching mode	frequency bucket + token type + token length
Matched repetitions	5
Random repetitions	5
Max length	160

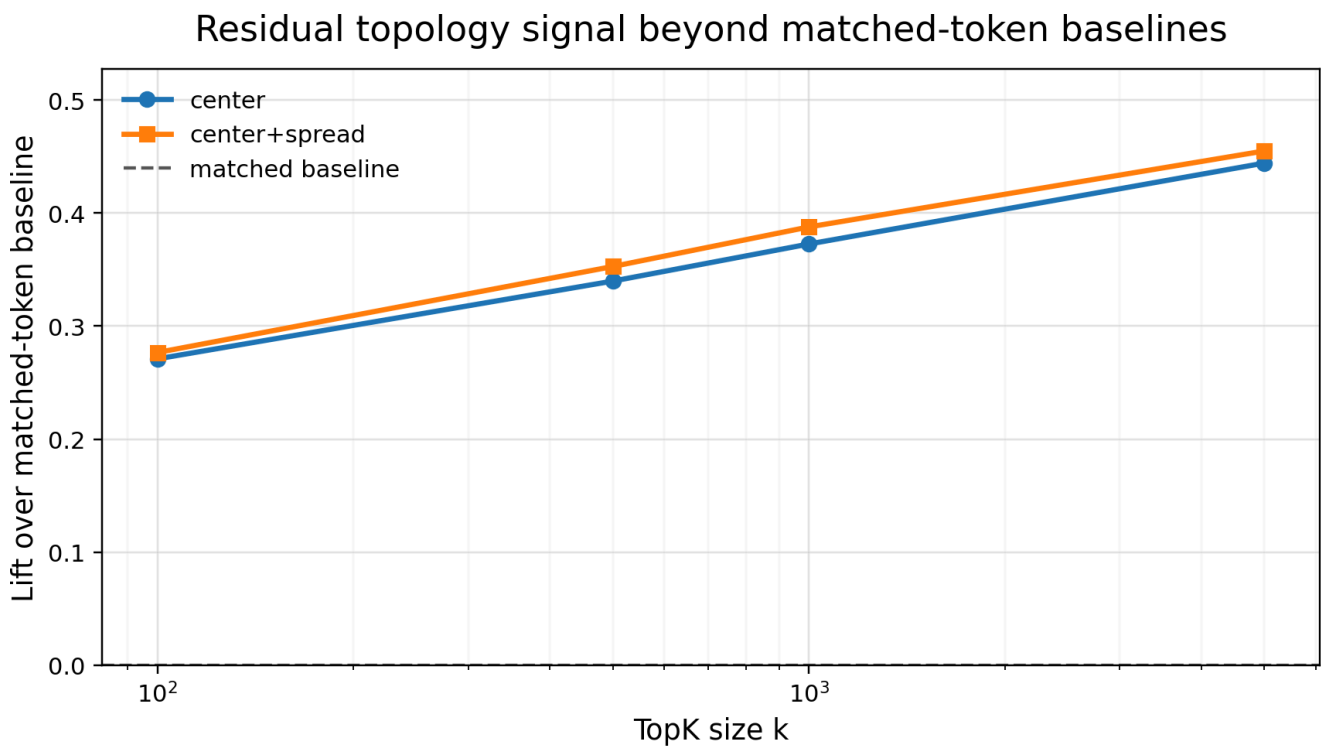
D.2 Summary by TopK and Signature

TopK	Mode	Real Topo	Matched Topo	Lift vs matched
100	center	0.3309	0.0599	0.2710
100	center+spread	0.3409	0.0643	0.2766
500	center	0.4664	0.1265	0.3398
500	center+spread	0.4851	0.1324	0.3527
1000	center	0.5312	0.1586	0.3726
1000	center+spread	0.5509	0.1633	0.3877
5000	center	0.6459	0.2017	0.4442
5000	center+spread	0.6574	0.2025	0.4549

Across all layer / TopK / mode combinations:

224/224 combinations have $\text{Lift}_{\text{matched}} > 0$.

D.3 Residual Lift Trend



Appendix Figure D.1. Earlier token-statistic controls remain useful as artifact checks. After PSG realignment, this figure should be read as evidence against simple token-frequency/type/length artifacts, not as evidence for semantic row-identity causality.

D.4 Interpretation

The matched baseline is stronger than the uniform random baseline, as expected. This shows frequency, token type, and token length do explain some vocabulary-space structure. However, the real TopK signal remains far above the matched baseline. The matched control therefore supports:

The identity-only TopK signal is not explained by coarse token frequency, type, or length artifacts.

It does not eliminate all vocabulary-space confounds, especially output-embedding anisotropy or deeper semantic-frequency structure.

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Appendix E. Corrected W -Disruption Control

E.1 Motivation

The initial reviewer-risk protocol asked whether the identity-only TopK effect depends on learned vocabulary geometry or on projection-selection artifacts. The first version exposed a serious risk. The corrected version was designed to remove two obvious issues: layer-0 degeneration and final punctuation pooling.

E.2 Corrected Configuration

The corrected audit uses Qwen2.5-1.5B-Instruct, 196 evaluation texts, last-non-punctuation pooling, and skips the embedding layer. It evaluates layers 1–28 with TopK values 100, 500, 1000, 5000, center and center+spread signatures, and 200 bootstrap replicates. Same-as-readout variants are retained only as sanity checks.

Primary controls:

- real W ;
- row-permuted readout + original W realization;
- random-Gaussian readout + original W realization;
- orthogonal-rotated readout + original W realization;
- hidden orthogonal rotation + real W ;
- fixed random tokens;
- centroid-norm-matched random tokens.

E.3 Main Result

For center signatures at $k = 5000$:

Control	Mean Topo	Mean Δ vs real	Mean 95% CI low	Mean 95% CI high
real W	0.493	+0.000	0.471	0.566

Control	Mean Topo	Mean Δ vs real	Mean 95% CI low	Mean 95% CI high
row-permuted readout + original W	0.619	+0.125	0.609	0.671
random-Gaussian readout + original W	0.628	+0.135	0.620	0.678
orthogonal-rotated readout + original W	0.630	+0.137	0.607	0.688
hidden orthogonal rotation + real W	0.683	+0.189	0.661	0.732
fixed random tokens	0.021	-0.472	0.035	0.115
matched centroid-norm random tokens	0.027	-0.467	0.040	0.121

For center+spread signatures at $k = 5000$:

Control	Mean Topo	Mean Δ vs real	Mean 95% CI low	Mean 95% CI high
real W	0.362	+0.000	0.357	0.437
row-permuted readout + original W	0.546	+0.184	0.537	0.607
random-Gaussian readout + original W	0.523	+0.161	0.509	0.585
orthogonal-rotated readout + original W	0.440	+0.079	0.428	0.513
hidden orthogonal rotation + real W	0.476	+0.115	0.457	0.548
fixed random tokens	0.022	-0.339	0.038	0.115
matched centroid-norm random tokens	0.028	-0.334	0.042	0.121

The random-token controls are low. The disrupted readout controls are high. This combination is the important part: the phenomenon is not explained by arbitrary token sets, but it also does not require the exact learned semantic pairing of W .

E.4 Interpretation

The audit fails to support the semantic- $\mathcal{W}$ claim. It supports a narrower object-level claim: identity-only TopK sets define a vocabulary-side geometric observable of hidden states. The geometry-preserving effect survives controls that disrupt the readout, so it must be studied as a projection-selection conditional-mean readback effect before it can be treated as a learned semantic mechanism.

This appendix is therefore not a minor robustness check. It changes the paper's claim boundary.

Appendix F. Projection-Selection Conditional-Mean Readback Series

F.1 Model-Pipeline Projection-Selection Decomposition Audit

The model-pipeline decomposition audit decomposes the readout pipeline inside the model setting. It shows that Gaussian score readouts can recover hidden geometry comparably to real $\mathcal{W}$, that TopK is far stronger than random-K, and that centroid aggregation is the dominant contributor. This motivates a pure high-dimensional null.

F.2 Pure Gaussian Selection Null

The pure Gaussian selection-null audit removes the model, tokenizer, vocabulary, and learned matrix. It constructs synthetic queries H_i , random candidate readout vectors R_j , selects TopK by $H_i R_j^\top$, and aggregates selected vectors by centroid. The pure Gaussian TopK centroid strongly preserves query geometry, while random-K does not.

The empirical mechanism is:

$$Z_i = \frac{1}{k} \sum_{j \in \text{TopK}(H_i R^\top)} R_j \approx \alpha H_i.$$

For BottomK:

$$Z_i^{\text{bottom}} = \frac{1}{k} \sum_{j \in \text{BottomK}(H_i R^\top)} R_j \approx -\alpha H_i.$$

F.3 Real-Readout Residual Audit

The real-readout residual audit compares real $\mathcal{W}$ against matched Gaussian and distributional nulls. It finds no positive topology-preservation residual of real $\mathcal{W}$ beyond matched projection-selection nulls. Row permutation does not reduce preservation. Whitening reduces real- $\mathcal{W}$ performance, suggesting that raw covariance or anisotropy modulates readback, but not through semantic row identity.

F.4 Conditional-Mean Readback Mechanism

The conditional-mean readback audit verifies the mechanism mathematically. For Gaussian candidate vectors and a unit query h , projection scores satisfy

$$s = h \cdot R \sim \mathcal{N}(0, 1).$$

TopK selection corresponds to conditioning on the upper tail:

$$s \geq t, \quad t = \Phi^{-1}(1 - k/V).$$

The selected centroid has conditional mean aligned with the query:

$$\mathbb{E}[R \mid h \cdot R \geq t] = \mathbb{E}[s \mid s \geq t]h = \frac{\phi(t)}{1 - \Phi(t)}h.$$

For sphere-normalized candidates, the leading scale is approximately

$$\alpha(d, V, k) \approx \frac{1}{\sqrt{d}} \frac{\phi(t)}{1 - \Phi(t)}.$$

Empirically, α_{emp} matches α_{theory} with very small error. The estimator remains noisy,

$$Z_i = \alpha(d, V, k)H_i + \epsilon_i,$$

but its self-alignment is high enough to preserve pairwise query geometry.

F.5 Final projection-selection geometry Interpretation

The projection-selection geometry series yields the following decomposition:

$$\text{ObservedTopKPreservation} = \text{projection} - \text{selectiongeometry readback} + \\ W \text{ distributional modulation} + \text{No detected positive semantic rowext} - \text{identity residual}.$$

This decomposition replaces the earlier semantic- W interpretation. The strongest supported claim is no longer that TopK preservation proves learned vocabulary semantics. It is that identity-only TopK sets are geometry-preserving readout-side observables, and that their preservation is primarily explained by projection-selection conditional-mean readback.

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